Supplementary Material for
Maintaining LLM-Agent Conversation History as a Prefix-Stable Streaming Materialized View

***Abstract*—This supplement collects, verbatim, the arm construction, offline control gates, derivations, reconciliation notes, per-number qualifications, audit tables and figure detail that the main paper points to. Every number here is the number that used to sit in the single-file appendix; the split moved text by line range and did not change a digit.**

S.A. Cache-field validation detail

The six checks summarized in Sec. VI-A, with the numbers. *(i) Token granularity.* 272 real prompts tokenized with a local Qwen BPE: the server's `prompt_eval_count` exceeds the proxy count by a *constant* 16 tokens on every one of them (the chat-template wrapper). For every consecutive-turn pair we compare the longest common *token* prefix of the two prompts to the cached count reported on the second: over 128 pairs per cell, the sliding cell reports mean token-LCP 29.6 vs. cached 32.6 and the stable cell 316.7 vs. 319.7, with zero under-reuse pairs (cached $< 0.9 \times$ LCP, which would mean BPE joins or eviction are breaking the mechanism) and zero over-reuse pairs beyond $+4$ tokens (which would mean stale nodes leaking across runs). The residual $+3$ is not noise and not a bias: re-reading the persisted event file, `cached` exceeds the raw-prompt token-LCP by *exactly* 3 tokens on every one of the 256 pairs that has a predecessor, and the only deviations are the 15 first-pair rows of a stream (LCP $= 0$ by construction, cached $= 18$) and one warm-up row. Three template tokens therefore precede the first prompt token inside the server's own tokenization, which is also why the 16-token constant offset splits as 3 leading plus 13 trailing template tokens. The comparison is thus anchored 3 tokens earlier on the raw-prompt side than on the server side, and the naive arm's 17.6-token hit (check v) is measured against that same anchor. The first two checks are recomputable from the persisted event file with no model call. *(ii) Constructed prefix.* Two prompts sharing an exactly-k-token prefix and differing only in the trailing question return cached $= 1679/1688/1720/1841$ for $k = 1670/1678/1710/1831$ (within 0.6%) and recompute $= 19$ tokens in all four cases. *(iii) Eviction.* Re-asking one warmed prefix after $0, 4, 16, 40$ interleaved unrelated requests leaves cached unchanged at 1693 of 1694 tokens. *(iv) Server.* `ollama` 0.33.3 on four GPUs (3× RTX 4090 24 GB, 1× L40 48 GB, driver 535.230.02); served model `qwen3:14b`, 14.8B parameters, Q4_K_M, native context 40960, with `num_ctx`= 8192 set per request by the client. The client sets neither `keep_alive` nor a cache TTL; both stay at server defaults. The CLI does not expose the bundled llama.cpp revision, so instead of guessing a commit the artifact pins the build by content: SHA-256 of the `ollama` binary ($7deaad14\ldots$), of the two llama libraries (`libllama.so` $af4d01b0\ldots$, `libllama-server-impl.so` $47558fc8\ldots$), and the registry digest, size, family and quantisation of all eight served models, including the embedding model. Anything that changes the tokenizer, the template or the cache implementation changes those digests, so the environment is specified rather than merely described. *(v) Nonce.* Its one visible effect is the naive arm's 17.6-token mean `cached`: the nonce plus the shared block header, which is exactly the reuse the mechanism predicts and no more. *(vi) Run-to-run stability of the latency field.* Two of the four GPUs carry other users' processes, so the same turns can be timed differently on different days. Re-firing 23 published cost turns from two sessions for both arms, three times over with a fresh nonce per repeat (276 calls), the mean prefill per turn moves by 9.1% of its own level on the naive arm (1141.1/1042.9/1043.2 ms) and 3.9% on the stable arm (207.5/199.6/199.7 ms), while mean recompute moves by 0.2 tokens on the naive arm and 0.0 on the stable arm. That is the contention story behind our currency choice: a run-to-run spread of $\sim$ 98 ms on the naive arm is larger than the session-cluster interval we report for the same field, so the latency intervals describe conversation-to-conversation variation, not measurement noise, and every cost claim is stated in recomputed tokens, which are repeat-exact. The script and its per-call log are `repeat_ms_variance.py` and `ms_repeat_variance_scored.jsonl` (138 calls per arm, 0 server errors, longest prompt 1732 tokens, 21.1% of `num_ctx`).

S.B. Statistical procedure in full

The resample count is fixed per analysis script and not tuned: 4000 draws for the 2×2 factorial and the pooled danger contrasts, 6000 for the cost, bounded-head, layout and position analyses, all with a fixed seed. Cost gaps are reported as session-cluster bootstrap intervals, the cluster being the unit the design randomizes over; pooled quality effects use a Mantel–Haenszel test with a question-cluster bootstrap risk difference, so a repeated item contributes one resampled unit rather than one per arm, and paired fidelity contrasts use the exact McNemar test on discordant turn-pairs. The pooled stratum set is the pre-specified primary analysis (declared at revision) as every served (model, temperature) cell with no capability or significance filter; a size-filtered

pool is reported only as a labelled subgroup analysis, so the pooling rule cannot be chosen after seeing which layers look favourable, and because the same model appears at two temperatures the statistic is also given over the mutually independent temperature-0 strata alone. Cross-model statements about effect *size* are tested against an external anchor (the checkpoint size class from the model card) with an exact permutation test at n=7 models rather than asserted from a table (`analyze_capability.py`). The contamination-hazard association is a logistic model on a semantic-proximity term with false-discovery-rate-adjusted p-values, and the sized control isolates mechanism from compactness.

S.C. SESOI ANCHORING AND EQUIVALENCE-TEST AUDIT

The two floors in Sec. VI-A are anchors on quantities a deployer would act on rather than conventional effect-size labels: 20 recomputed tokens is 1.3% of the naive arm's measured per-turn prompt (1574.9 tokens, Table II), and the fidelity floor is set in the range where a substitution-rate shift changes which arm a system builder would pick. Both were declared before the confirmatory runs and not revised afterwards. The equality audit resamples sessions, the unit the design randomizes over, with the resample count fixed a priori, using 6000 draws and one-sided tails evaluated against ± 0.03 and ± 20 [1]. The test was added to address the joint-versus-decoupled reading, so it is a re-analysis of the frozen run files at the already-declared margins, not a new outcome.

S.D. DATA PROVENANCE: WHICH LONGBENCH MATERIAL, AND WHAT IS CONSTRUCTED

The two LongBench uses in this paper are different materials and should not be conflated. The passage-danger slice takes LongBench's `hotpotqa` multi-document QA items directly, question, candidate passages and gold answer included: the release's held-out `hotpotqa.jsonl`, `2wikimqa.jsonl` and `musique.jsonl` test files, used with no division of our own, giving 300 usable questions (68 hotpotqa, 131 2wikimqa, 101 musique; K=4 distractors, truncation 500 tokens [2]), whereas the multi-turn harnesses (cost, 2 × 2, layout, position stress) take only the *sentence bodies* of `hotpotqa`, `2wikimqa` and `musique` and reassemble them into synthetic append-only sessions, so a session is topically coherent real prose but its turn sequence, gold placement and near-twin values are constructed. Semantic proximity for the contamination-hazard model and the danger-slice thresholding uses bge-m3 cosine. The nonce requirement came from a smoke-test failure: without per-config nonces the sliding window fakes a whole-prompt cache hit off a prior invocation's radix node, a cross-run contamination that would inflate the naive arm.

S.E. RUN INVENTORY, SAMPLE FLOW AND MAINTENANCE LEDGER

Selection rules behind Sec. VI-A. The passage slice keeps the questions whose *constructed* 500-token naive window retains a near-adversary passage at bge-m3 cosine ≥ 0.60. An embedding-only prevalence scan over 300 usable questions fixed that yield at 47 paired questions per stratum before any call was issued (Table S8 reports the count at every threshold), and the served pool per stratum is what the 4,664-call passage-danger row of Table S1 multiplies out. Three pools were served, and which stratum got which is a fact about the run files, not a post-hoc description: six strata took the full 300-question scan pool (`llama3`/`mistral`/`qwen2.5:14b`/`qwen3:14b` at t=0, and both `qwen3:8b` strata), six took the 74-question ≥ 0.55 pool (both `gemma2:2b` strata and the four remaining t=0.6 strata), and two took a 44-question ≥ 0.60 pool, namely the 47 danger questions minus three `2wikimqa` items that the two `qwen3.8:27b` strata (t=0, t=0.5) were never served. The three absent items are recorded rather than explained, and both counts are readable off the trial files: the ≥ 0.60 subset of the scan pool is 47 questions (14 `hotpotqa`, 25 `2wikimqa`, 8 `musique`, counted on the `length_only` arm, whose retained passages the prevalence scan selects on), and the shorter file those two runs read carries 44 of them (14/22/8), the three missing ones being `2wikimqa`. That is $6{\times}2{\times}300 + 6{\times}2{\times}74 + 2{\times}2{\times}44 = 4{,}664$ calls in 14 files, of which the primary ≥ 0.60 slice is $6{\times}47{+}6{\times}47{+}2{\times}44 = 652$ paired questions (1,304 calls) carrying the C4 contrasts, the declared ≥ 0.55 sensitivity adds 324 pairs (648 calls), and 3,360 calls were served on questions outside the primary slice; they enter the prevalence table and the audit, not the contrasts. The arithmetic is printed by `sample_flow.py` off the run files. The single-field null screens 345 groups down to 91 eligible pairs per arm. Table S4 traces every grid and stream from generated records to analysed cells, per arm, computed from the run files rather than typed (`sample_flow.py`); Table S5 is the operator's own cost ledger.

Flow accounting. The $720 \to 552$ and $480 \to 408$ differences a reader will notice against Table II and Table V are the pre-declared windows (turns 7–29 of 30; turns 3–19 of 20), not dropout, so the paired denominators are equal by construction and no arm-specific survivorship is possible. The overflow worry that goes with this design is measured in the same file (the naive arm of the cost legs peaks at 1850 tokens, 1864 on the second model, and the highest prompt anywhere in the audit is 1870 (the competitor leg's naive-window control on the second architecture) against the pinned `num_ctx`= 8192: 22.8%).

Leg shapes and the three legs itemised in the audit. Three legs are named here rather than only in their own sections: the rewritable-index fidelity leg, 4,320 calls in 2 files; the index cost leg, 2,300 calls in 1 file; and the external content-rule leg of Sec. VI-B2, 7,728 calls in 2 files, whose 7 arms are the compressed roster of 4 open keep-rates and 2 gold-protected rates plus the same-run length-only control. The audit also states the shape of each leg, because the grid size and the call count differ and a reader is entitled to know why: the index fidelity leg is one 2,160-trial grid (three arms ×720 cells) served whole to each of two architectures, which is why the deterministic gate quotes 2,160 trials while 4,320 calls were

TABLE S1

MODELS SERVED AND INFERENCE CALLS PER EXPERIMENT, SUMMED OVER THE 38 CONFIRMATORY SCORED RUN FILES (THE 34,134 CALLS AUDITED IN SEC. VI-A); THE LAST ROW IS THE LATENCY-REPLICATION DIAGNOSTIC OF CHECK VI, COUNTED IN THE 34,410-CALL 39-FILE AUDIT BUT SUPPORTING NO CLAIM OF ITS OWN. ARCHITECTURES ARE AS REPORTED BY `OLLAMA SHOW`; THE SEVEN CHECKPOINTS SPAN *five* DISTINCT REPORTED ARCHITECTURES BECAUSE `MISTRAL:LATEST` IS REPORTED AS `LLAMA` AND BOTH `QWEN3` TAGS AS `QWEN3`. `QWEN3.8:27B` IS A LOCAL 27.3B Q4_K_M TAG WHOSE REPORTED ARCHITECTURE IS `QWEN35` (A VISION-CAPABLE CHECKPOINT; ONLY ITS TEXT PATH IS EXERCISED); THE OTHER SIX ARE STOCK TAGS, AND `LLAMA3:LATEST` IS THE 8B LLAMA-3 CHECKPOINT. THE THREE PASSAGE-DANGER ROWS ARE THE POOLS ACTUALLY SERVED: 6 STRATA ON THE 300-QUESTION SCAN POOL (`LLAMA3`, `MISTRAL`, `QWEN2.5:14B`, `QWEN3:14B` AT t=0 AND BOTH `QWEN3:8B` STRATA), 6 ON THE 74-QUESTION ≥ 0.55 POOL (BOTH `GEMMA2:2B` STRATA AND THE FOUR REMAINING t=0.6 STRATA), 2 ON A 44-QUESTION POOL (`QWEN3.8:27B`); THE MULTIPLICATION IS SHOWN IN SUPPLEMENTARY SEC. S.E.

Experiment	Model (architecture)	Calls
Cost, primary (3 arms $\times$552)	qwen3:14b (qwen3)	1656
Cost, 2nd model (3 arms $\times$552)	qwen2.5:14b (qwen2)	1656
Bounded head (3 arms $\times$396)	qwen3:14b	1188
Joint 2 $\times$ 2 (4 cfg $\times$408)	qwen3:14b	1632
Layout 2 $\times$ 2 (4 cells $\times$408)	qwen3:14b	1632
Tail stress, forced commit	qwen3:14b, epochs 1–2	960
original (unforced) variant	qwen3:14b	480
2nd architecture	llama3:latest (llama)	480
Gold-position stress	qwen3:14b	720
2nd architecture	llama3:latest (llama)	720
Index fidelity (3 arms)	qwen3:14b	2160
2nd architecture	llama3:latest (llama)	2160
Index cost (5 arms)	qwen3:14b	2300
Shared-key index, re-served (2 arms)	qwen3:14b	1440
2nd architecture	llama3:latest (llama)	1440
Content rule, primary (7 arms $\times$552)	qwen3:14b	3864
2nd architecture	qwen2.5:14b	3864
Passage danger, scan pool	6 strata, 2 arms $\times$300 q.	3600
served at ≥ 0.55	6 strata, 2 arms $\times$74 q.	888
27B pool	2 strata, 2 arms $\times$44 q.	176
Field-level danger, paired	qwen2.5:14b, qwen3:14b/8b	728
Field-level 3-arm policy	qwen3:8b	270
Concurrency probe	qwen3:14b	120
Total (38 confirmatory files)	7 ckpts, 5 archs	34134
diag.: latency replication	qwen3:14b	276

issued; the gold-position leg is the same shape, one 720-trial grid served twice; the content-rule leg is one 552-trial grid (the published cost sessions on their own pre-declared window) served whole to each of two architectures at 7 arms per trial. Those numbers and the passage-danger multiplication above are printed by `sample_flow.py` off the run files, not typed in.

a) Maintenance cost ledger: The operator's whole ledger is Table S5: a fold is a string format taking ≈ 0.15 ms (append-only case) to < 1 μs (roll-up case) per turn, with 0 model calls and 0 output tokens. Its real price is on the recompute side: the capped head's roll-up rewrites leading bytes, so the 48 turns on which it fires recompute 209 tokens more than an append turn, i.e. 25.3 tokens amortized per turn, 2% of the 1293-token per-turn saving the bounded policy measures. The currency is R, not B: on such a turn the prompt *falls* (six capped head lines become one) and what rises is the re-prefill. Every recompute figure in the paper is read from the served request, so it is *after* maintenance, not before it.

S.F. ADDITIONAL MEASURED DETAIL

Four tables carry the statistics the body quotes in a clause each: the bootstrap that attributes the cost gap to invalidation rather than to compactness (Table S6, behind Sec. VI-B1); the factorial main effects and their interaction on both axes (Table S7, behind Sec. VI-C); the number of questions that clear each danger-slice threshold, which is how the operating point avoids being cherry-picked (Table S8, behind Sec. VI-C5); and the logistic refit that puts contamination risk in order of semantic proximity (Table S9, behind its hazard paragraph in Sec. VI-B1). The **C** numbers in the captions are the ledger rows of Table S15, and every value here comes from a run file named in Table S2.

A. Per-stratum danger-slice detail

The body states the quality benefit as one pooled reading, so the strata behind it are shown first. Table S10 gives substitution under each arm per (model, temperature) stratum, with Wilson intervals and the McNemar discordance counts the pooled statistics are built from; Table S11 keeps the primary pool, the $\geq$ 7B sensitivity pool and the withdrawn outcome-filtered subgroup side by side; Table S12 is the external capability anchor that retired the size-dependence reading; and Table S13 is the single-field slice where the same contrast is exactly null. Together they set the scope of the claim: direction in 12 of 14 strata, magnitude neither ordered by model size nor pinned beyond ~ 0.03, and no hazard at all once a query can only have one current value.

S.G. CAPABILITY ANCHOR: THE READING WE DROPPED

We previously read Table S10 as a capability dependence, i.e. that benefit *size* tracks capability rather than architecture. Tested against an external anchor, that reading does not survive and we drop it. Table S12 gives the per-model temperature-0 risk differences over size classes 2–27B and the three checks we ran on them (`analyze_capability.py`): the spread is not statistically distinguishable from sampling noise, the rank correlation with the checkpoint size class is near zero and its sign is decided by a single model under leave-one-out, and the ordering has counter-examples on its face (`mistral` 7B above `qwen2.5` 14B, `qwen3.8:27b` below both 14B checkpoints). Those are the checks behind the weaker reading stated in Sec. VI-C5, which is the one the paper claims.

S.H. LAYOUT-ONLY FACTORIAL AT FIXED CONTENT

Table S14 is the control behind Sec. VI-C: labels crossed with prefix stability on identical content, with the front-reuse separation verified offline.

S.I. CLAIM LEDGER AND TABLE-READING NOTES

Table S15 states each claim, the single number that carries it, the evidence grade that sets its wording strength, and the observation that would falsify it. It is the contract between the prose and the data: a claim is written no stronger than its grade, and each is checkable against a named experiment in Table S3.

a) Why two rows of one model can share a marginal rate: Each model is served twice, at two temperatures over the *same* question set, so identical marginal rates in Table S10 (`llama3` 0.213 / 0.128 at both t=0 and t=0.6) are not duplicated rows: the calls are separate records, and only 41–43 of 47 outcome pairs coincide for the models rerun at a different seed (1234 at t=0, 777 at the elevated temperature), while prompt tokens are identical by design and sampled answers differ. Two exceptions are stated rather than hidden: `qwen3:8b` (t=0 vs t=0.7) and `qwen3.8:27b` (t=0 vs t=0.5) reuse seed 1234, and the former's 47 outcome pairs agree exactly, so its second stratum carries no independent information. Every pooled statistic is therefore reported twice, once over all 14 cells and once over the 7 mutually independent temperature-0 models.

b) What the `corr` column does and does not say: One cell in Table V invites the wrong reading: correctness falls from 0.696 (`c01`) to 0.512 (`c11`) even though both arms score 0.000–0.002 substitution, and the paired contrast is far from noise (exact McNemar on the 408 shared turn-cells: b=99, c=24, $p = 5.2 \times 10^{-12}$). Sorting every answer into four disjoint categories (record id quoted; a span of ≥ 3 shared 4-word shingles with the *gold* record's body; likewise with a *distractor*'s body; neither; computed by a script over the same run file, so the split is auditable rather than eyeballed) shows what moved. Adding the semantic lever shifts answers from the first category to the second, id-quoting $0.750 \rightarrow 0.512$ and gold-content quoting $0.203 \rightarrow 0.453$, while the distractor-content category stays at $0.002 \rightarrow 0.000$ and the substitution contrast is b=1, c=0. The first of those two rates is not the table's `corr` column for the same cell, and the difference is the definition rather than a second measurement: the categoriser scores an answer as id-quoting whenever the gold id occurs in it, while `corr` also requires that no competing value occurs, so every `corr`-correct cell is id-quoting and the extra cells are answers that list the gold record *and* a competitor. That is where the 0.054 gap lives (22 of 408 cells in `c01`, the same construction that gives the naive cell its 0.368 id rate against 0.118 `corr`); the two agree exactly in `c10` and `c11`, where no answer names both. In other words the arm that loses `corr` is increasingly answering with the gold record's own text (e.g. replying "*The Real Glory*" where the harness expects the id `R20` that labels that record) rather than with a competitor's. The naive cell behaves the opposite way (id-quoting 0.368, distractor-content 0.282), which is the failure the metric was built to catch. `corr` in this table is therefore a verbatim-format match, and the 0.184 drop is an answer-style effect we do not interpret as a fidelity loss; the fidelity claims in this paper ride on substitution and on the forced-commit protocol of Fig. 4, where the response space is a single value and format has nowhere to go.

The same dissection, run over the index grid, is what makes that distinction concrete in the other direction. Table S16 sorts all 720 answers per arm per architecture into the same buckets. With the distractors re-keyed, no answer in any of the three arms names a competing record: the *another record* and *substitution* columns are 0.000 in every one of the six index-leg rows, against 0.449 and 0.158 substitution in the published shared-key cells of the last row of each block. What `plain` still loses is gold content restated in prose without its id (0.237 of qwen3's answers, against 0.069 under `index`), so the index's headline correctness gain ($0.753 \rightarrow 0.928$ on qwen3, $0.981 \rightarrow 1.000$ on llama3[1]) is the format effect this supplement warns about, not a measured drop in substitutions. The split is POST-HOC and exploratory: it was computed after the primary contrasts, no arm assignment or pre-declared comparison depends on it, and it is reported to locate the residual loss rather than to support a claim.

S.J. Results narrative in full

This section gives the per-leg narrative behind the results reported in the body (Sec. VI-B1). Every measured number, interval and scope caveat in the body appears here as well. The floats the body keeps (Tables II, IV, S19, S21, III) are referenced here rather than duplicated.

A. Cost: prefix stability, not compactness

At matched answer accuracy, prefix-stable maintenance cuts per-turn recompute by 6.26× (95% CI $[6.22, 6.31]$: 1557 to 249 tokens; the ratio is formed on the unrounded paired means, so it reads 6.26 rather than the $1557/249 = 6.253$ implied by the rounded table entries) and prefill latency by about 5× (1056 to 205 ms) relative to naive length-based truncation, and we demonstrate with a size-matched pinned control that recovers 97% of the gap that the source of the gain is prefix invalidation rather than sending fewer tokens. The same ordering holds on `qwen2.5:14b`: the paired recompute *gap* reproduces to within 0.1 token (1308.6 vs. 1308.7 tokens/turn) even though the absolute levels differ with the tokenizer ($1549 \rightarrow 241$ rather than $1557 \rightarrow 249$), and the layout-to-cost relation also holds on a second family, `llama3`, on the forced-commit 2×2 arms rather than the cost harness (Fig. 4). Fig. 3 shows the three arms and Table II the raw counters; the paired bootstrap contrasts are in Table S6. The pinned control is the load-bearing piece: it carries the *same* total tokens as the naive window yet reproduces $1272/1309$ of the recompute gap, so the gain cannot be explained by sending less. The residual runs in our favour and we state it in the measured direction: the frozen-pin control recomputes 36.5 tokens/turn (CI $[35.2, 37.8]$) *more* than M-A (285.2 vs. 248.7 mean), 2.8% of the naive→M-A gap; the control carries the verbatim block where M-A carries a digest, and we do not decompose the residual further. But M-A also buys bounded carried memory (445.5 vs. 1584.3 tokens) that the frozen control does not, so the two sit on different axes and we report both. (The recompute column is the mean difference of the two mean counters, $445.5 - 196.8 = 248.7$, which is what rounds to 249 in Table II.)

[1]These `corr` values pool the two head lengths and three slots, so they differ from any single cell of Table S19.

B. When naive truncation collapses

Naive recency-window truncation is fragile when it retains competing near-twin values, and we find that semantic proximity to the gold predicts contamination risk. On the three-arm natural field run (270 trials, 13 contamination events, `qwen3:8b`) a logistic model contamination $\sim$ max_sim $+K$ gives a positive proximity term (unpenalised $\beta = 8.16$, false-discovery-adjusted $p = 0.001$; Table S9); because all 13 events fall above the τ_{near} band the raw coefficient is inflated by near-separation, so we read the effect from its monotone direction and ROC AUC (0.74 penalised to 0.81 unpenalised) rather than the coefficient magnitude. This run is distinct from the single-field paired slice (91 eligible pairs/arm, all-zero discordance, Table S13): the null is on the single-field slice, the hazard on the three-arm field where max_sim spans 0.22–0.74, and the two are not in conflict. Substitution for the naive arm reaches 0.392 (CI $[0.346, 0.440]$) in the base 2×2, 0.808 $[0.729, 0.869]$ under the tail-distractor stress, and 0.55–0.97 under the forced-commit protocol, where it commits to a correct value on only 0.03–0.45 of trials against 0.99–1.00 for the gold-keeping arms (Table V, Fig. 4). The forced-commit range spans two epochs on `qwen3:14b` (0.617 and 0.550 substitution, 0.383 and 0.450 correct) and one on a second family, `llama3:latest` (0.967 substitution, 0.033 correct); the naive-versus-stable contrast is exact McNemar b=74/c=0, b=66/c=0, b=116/c=0 respectively, $p < 10^{-19}$ in all three. The second family is therefore *more* fragile than the first, not less, and we report the spread instead of a single-family headline. The `llama3` column is read from the run whose 480 rows all log `llama3:latest`, and the two `qwen3` columns are the two passes of one `qwen3:14b` run file, kept as separate epochs; every run file in Table S2 carries exactly one model tag, checked row by row. Table V reports the same ordering. On easy single-field retrieval the same hazard is close to absent ($b = c = 0$ discordant pairs in every configuration, Table S13), so we scope the hazard claim to layout-dependent, near-twin-rich regimes rather than to context length per se.

C. Which layout factor suppresses substitution

That a *cost*-motivated layout also improves fidelity was not assumed, but the 2×2 above does not say *why*: the stable arm changes two things at once, the ordering that keeps its leading bytes fixed and the section labels that name the blocks. We crossed them on identical content (Table S14 in Supplementary Sec. S.H): d=0 throughout, gold and competing near-twins rendered verbatim in every cell, 12 sessions $\times 34$ mature turns $\times 4$ cells $= 1632$ calls, 0 server errors. The factors are separated by construction, verified without any model call: front reuse (longest common prefix between consecutive-turn prompts) is 2.0% and 7.1% in the sliding cells versus 51.2% and 51.3% in the stable cells, so the label factor moves no bytes at the front; competing values present in context are 1.06 in all four cells; total prompt tokens spread by 1.97%, which is the header-length asymmetry itself ($\approx$12 tokens) and is reported rather than hidden.

The two axes then split cleanly. Stripping the labels from the stable arm leaves substitution at $0/408$ (Wilson upper bound 0.009): the labels are not what suppresses contamination. Adding them to the sliding arm moves substitution from 0.473 to 0.387 (exact McNemar b=59, c=94, $p = 0.006$), still a 0.387 residual against the stable arm's $0/408$ (Wilson upper bound 0.009), so labelling helps only where the front is already sliding. (We quote the residual as an absolute gap rather than a ratio: the stable-arm rate is 0, so any multiple-of-0 statement would be undefined.) On cost the label contrast is $+6.1$ tokens/turn against stability contrasts of -260.4 (95% CI $[-273.0, -247.7]$) with labels on and -268.9 ($[-282.2, -255.6]$) with labels off. A cluster-robust linear-probability model on substitution (SEs clustered on the 12 sessions; the logit is not estimable because both stable cells are 0/408 events) assigns -0.473 ($p = 1.3\times10^{-8}$) to stability and -0.086 ($p = 0.081$) to labels, with an interaction of the same magnitude and opposite sign: the label term is entirely absorbed by the stable layout, and only the stability effect clears the pre-declared Δ substitution $\geq$0.03 floor.

What survives is a *placement* claim, not a wording claim: the accumulated record is kept as an older-first sequence whose leading bytes never move, while the freshly-slid competitor is confined to a separate volatile region, and that separation is what the model reads. The reading this zero first invites attributes it to a "structural authority cue" provided by the labelled block; the factorial does not support that attribution and we drop it. Two consequences follow. The fidelity and cost axes are not two names for one mechanism: they share the stability factor but not its cause, since cost responds to fixed bytes and fidelity to the ordering that makes the accumulated value the older, non-overwritten one. And this is why the semantic-admission lever has nothing left to add on the fidelity axis (next subsection): layout, not admission, closes the substitution gap at this operating point.

Two loose ends we report rather than smooth over. Correct answers are *lower* in the labelled stable cell (0.608) than in the unlabelled one (0.828), entirely in a third outcome bucket (0.392 vs 0.172) that names neither the gold nor a competitor: the labelled phrasing invites more restating and non-committal answers, and we claim nothing about labels being style-neutral. And this cell re-runs our stable arm in a second epoch: at `temperature=0` with a fixed `seed`, only 56% of its 408 answer strings are byte-identical to the main run's `c01` arm while substitution agrees on $407/408$. Short greedy outputs are reproducible here, hedging length is not (batching-sensitive floating point), so we anchor cost claims on the recompute counters and fidelity claims on substitution, which replicated.

Table IV gives the answer, and it is the opposite of immunity. With the gold at the head's front, substitution is 0.000 (Wilson upper 0.031) at both head lengths and both models. Move the same line to the middle of the same head and substitution becomes 0.550 (H=6) and 0.742 (H=24) on `qwen3:14b`, and 0.158/0.325 on `llama3:latest`; correctness collapses from 0.942/0.933 to 0.042/0.008 and

from 0.375/0.475 to 0.008/0.000. The paired contrasts are the largest fidelity effects we have measured on this platform: middle-first at H=24 is -0.925 (95% CI $[-0.967, -0.875]$, exact McNemar $0/111$ discordant, $p = 8\times10^{-34}$) on `qwen3:14b` and -0.475 ($[-0.567, -0.383]$, $0/57$, $p = 1\times10^{-17}$) on `llama3:latest`, with substitution counterparts $+0.742$ and $+0.325$; every planned contrast has the same sign on both architectures, and the worst loss is $30\times$ our pre-declared 0.03 floor. Two shapes matter for the reading. The gradient is front-dominant and *monotone*, not U-shaped: middle and back slots are both far behind the front and are not ordered consistently against each other (H=6 middle−last on substitution -0.108, p=0.047 versus -0.075, p=0.093), so we claim a primacy gradient inside the head, not a recency recovery. Head length costs nothing when the target is at the front (H=24−H=6 at front: correct -0.008, CI $[-0.042, +0.025]$, p=1; $+0.100$, p=0.081) but deepens the deficit when it is not (at middle, substitution $+0.192$ $[+0.083, +0.300]$, p=0.001 and $+0.167$ $[+0.083, +0.250]$, p=0.0003; a cluster-robust LPM puts the middle×length interaction at $+0.192$, p=0.0007). Position is thus actionable on the fidelity axis in exactly the regime the earlier table left untested.

What this does to our own claim is narrow it, and we say so in the claim ledger: the stable-head arm's 0.000 substitution is a *region-separation* result (the probed value sits in the volatile tail and the head is not asked about), not evidence that an append-only head is immune to position. Inside the head the front line dominates, so an older-first head confers authority on the *oldest* surviving record. Our view semantics are consistent with that only because the current binding of a key is what a query asks for and it lives in the tail; a workload that recalls a *historical* binding out of the head faces this gradient directly, and the controller's only handle on it inside a single append-only view is the expensive one (place that record at the head front, i.e. break the prefix, or surface it in the tail). That motivates a second view rather than a better score: Sec. VI-D tests whether a bounded, rewritable block of current bindings is the cheap handle this finding asks for, and Sec. VI-D prices it. Head-internal position is therefore a third layout axis with a measured fidelity price, whose remedy, if the remedy works, is a different view, not a different eviction order.

D. A rewritable binding view, adopted and tested

That last sentence is the weakest position in the paper: if the only handle on a measured fidelity failure is an operation that forfeits the cached prefix, the maintenance model is missing a view. Sec. V-B sketches the missing view as a mechanism argument. We adopted that alternative and test it, because a proposal we merely cite would be the thinnest kind of contribution in a paper about view maintenance.

a) Result: Table S19 gives the grid: no server errors, no exclusion, 120 sessions per cell. The reading is uneven between the two architectures, and we report it that way rather than pooling the cells that agree.

P1 holds on qwen3 and fails on llama3. In the re-keyed `plain` arm the gradient is present at every contrast on qwen3: middle−first -0.217 [-0.292,-0.142], $p = 2.2\times10^{-7}$ at H=6 and -0.375 [-0.467,-0.283], $p = 6.8\times10^{-13}$ at H=24; last−first -0.250 and -0.492. On llama3 the same arm sits between 0.967 and 1.000 in five of its six cells, and only the H=24 last−first contrast clears the floor (-0.050, $p = 0.031$). The pre-declared consequence applies to the second architecture: this arm has almost nothing there for the block to fix, so its `index` numbers are a ceiling comparison.

P2 fails as written on qwen3: the block attenuates the gradient, it does not flatten it. The within-`index` contrast is -0.050 [-0.117,+0.017] at H=6, inside the floor, but -0.158 [-0.233,-0.083], $p = 6.6\times10^{-5}$ at H=24, outside it. The block *removes* 77% of the short-head gradient and 58% of the long-head one; what survives in the arm that carries the index is -0.050 and -0.158, and the latter is what fails P2. The estimand fixed for this purpose says the block moves the gradient rather than only shifting the level: the difference-in-differences is $+0.167$ [+0.092,+0.250], $p = 8.8\times10^{-5}$ at H=6 and $+0.217$ [+0.133,+0.300], $p = 4.3\times10^{-6}$ at H=24. On llama3 the `index` arm is 1.000 in all six cells (contrast $+0.000$, no discordant pair), which satisfies P2 in the uninteresting way P1's failure permits; its DiD is $+0.008$ ($p = 0.317$) and $+0.033$ [+0.008,+0.067] ($p = 0.046$).

P3 holds where there is a gradient to carry. `index_index_nok` at the disadvantaged slots is $+0.117$ [+0.050,+0.183], $p = 0.0013$ and $+0.233$ [+0.158,+0.308], $p = 7.5\times10^{-9}$ at H=6 middle and back, and $+0.175$ [+0.108,+0.250], $p = 5.7\times10^{-6}$ and $+0.400$ [+0.317,+0.483], $p = 7.1\times10^{-15}$ at H=24. The placebo keeps the gradient it was meant to keep (middle−first -0.150, $p = 2.8\times10^{-4}$; -0.333, $p = 4.6\times10^{-10}$), so the gain rides on the answer row and not on a second block sitting near the query. On llama3 only H=24 middle is beyond noise ($+0.075$ [+0.033,+0.125], $p = 0.0039$); the other three contrasts are $+0.017$ on two discordant pairs each. One regression per architecture says the same thing: over the `plain`@front@short reference, middle×`index` is $+0.192$ [+0.131,+0.252] and back×`index` $+0.408$ [+0.331,+0.485] on qwen3, against $+0.054$ and $+0.100$ for the placebo.

What the block does not buy. At the front slot, where arrival had already put the answer, the index arm is not better: -0.033 [-0.067,-0.008] at H=6, -0.017 [-0.058,+0.025] at H=24, against `plain`. The H=6 loss is the size of the pre-declared 0.03 SESOI and its cluster interval stays below zero, but four discordant pairs give the exact test no power ($0/4$, p=0.125; $2/4$, p=0.688 at H=24), so we read it as a cost at the floor, not as a demonstrated harm. Inside the index arm the ordering also inverts: last−first is $+0.033$ [-0.017,+0.083] at H=6 and $+0.042$ [+0.008,+0.083] at H=24, against -0.250 and -0.492 in `plain`, and the back slot beats `plain` by $+0.250$ [+0.175,+0.325] and $+0.517$ [+0.425,+0.608]. The block therefore redistributes correctness across slots (front held or slightly spent, middle and back recovered) instead of raising the level everywhere. And substitution is 0.000 in

every cell of both architectures, against 0.550–0.742 in the published shared-key stress (Table IV). That is what the re-key buys, and it locates C7 more precisely than Table IV alone does: the collapse reported there was the shared-key hazard and a position effect together, and the position effect by itself, the object of this section, is the $-0.217/-0.375$ gradient above. A post-hoc split of every answer by form (Table S16, Supplementary Sec. S.I.00b) shows where the remaining loss lands: in the re-keyed `plain` arm, 0.237 of answers restate the gold record in prose without echoing its identifier and only 0.010 name neither record, against 0.069 and 0.003 in the `index` arm. On this metric the block mostly repairs the report of the identifier, not the retrieval of the row.

b) Does the gain survive the hazard the re-key removed?: The reading above describes the re-keyed grid, and it is fair to ask whether the block earns its correctness only because the distractor was re-keyed away. We served the same current-binding `INDEX` back on top of the published shared-key `STATE` (gold and distractor both keyed `TARGET`, so a stale value genuinely competes under one key; the `plain` arm is the byte-identical C7 cell the model fails; 120 sessions per cell, no server error, no exclusion). Table S20 reports it, and the recovery is not a re-keying artifact. A correct index replaces the collapse where `plain` substitutes: index$-$plain correctness at the middle and back slots runs $+0.433$ to $+0.525$ on qwen3 (mean $+0.479$, every contrast $p \leq 3.1\times10^{-12}$) and $+0.592$ to $+0.825$ on llama3 (mean $+0.662$, every contrast $p \leq 10^{-21}$). The value in the row, not the block's presence, carries it: index$-$index_nok correctness averages $+0.632$ and $+0.554$, and an `index_nok` that names the competitor drives substitution *above* the raw state beneath it (mean index_nok$-$plain substitution $+0.217$ qwen3, $+0.590$ llama3). The model reads the view over the line it stands on: at the qwen3 front slot, where `plain` already sits at 0.942, a stale index pulls correctness to 0.058 ($0/106$ discordant pairs favoring the block, $p = 2.5\times10^{-32}$) and lifts substitution to 0.817. Two limits we keep rather than smooth over. On qwen3 the recovered mass leaves substitution but not every answer becomes a correct echo: a share of the collapsed cells turn non-committal (`other` up to 0.525 at $H{=}24$ middle), so the primary claim is elimination of substitution, not only a rise in correctness. On llama3 a within-index position gradient survives the hazard (`index` correctness 1.000 at the front, 0.833 and 0.592 at the $H{=}6$ middle and back), so under conflict the block attenuates the position effect rather than removing it. The consequence for C7 is the one Sec. VI-D now states: the collapse reported there was shared-key hazard and position together, and a maintained index answers the hazard even when the key is shared, so hazard suppression and fidelity recovery are the same measurement rather than a re-keying artifact.

E. An external content rule, measured rather than cited

The objection this section answers is that every comparison so far is *internal* to our own policy space: pinned, layout-only, position-permuted, the index arm. None of them is a published policy that rewrites the context, and arguing orthogonality without running the orthogonal thing is the thinnest kind of contribution. So we ran one.

a) Result: the content rule and the prefix axis move independently, in both directions: Table III gives the three coordinates. *Budget* (C8-P1): three arms sit at or below the view's 445.5 tokens (`comp_open20` 434.2, `comp_open12` 291.7, `comp_keep20` 442.6) so the comparison is size-matched where it needs to be. *Fidelity* (C8-P2): at the view's own budget, the unprotected pruner keeps the binding in 39 of 552 contexts and is correct in those same 39 (0.071 vs. the naive window's 1.000; paired -0.929 [-0.947,-0.909], exact McNemar $p = 7 \times 10^{-155}$ over 513 discordant pairs, none the other way). The attribution is exact and checked cell by cell: at every open rate the set of cells answered correctly *is* the set in which the binding survived ($551/551$, $338/338$, $39/39$, $0/0$; no cell is right without the row, no cell holding the row is wrong). The loss is the deleted row, not attention to it. The protected arm confirms this from the other side: the row survives all 552 prompts and correctness is 0.998 vs. 1.000, paired -0.002 [-0.005,0.000], a single discordant cell, inside the floor. *Recompute* (C8-P3): as pre-declared, every pruned arm recomputes fewer absolute tokens than the naive window (-597.1 to $-1{,}289.4$, all intervals below zero), which is the confound Amendment 1 named, not the mechanism. The mechanism is the share, taken on unrounded means: R/B is 0.919 at the low end (267.9 over 291.7, `comp_open12`) and 0.972 at the high end (960.2 over 987.6, `comp_open50`) for every compressed arm, 0.989 for the naive window and 0.558 for the view. Pruning to a fifth of the words does not buy reuse; it buys a smaller prompt that is still almost entirely recomputed. At the matched cell (`comp_keep20`, paired $\Delta B = -2.9$ [-8.7,+2.8], i.e. the same budget) the view recomputes 170.1 [165.9,173.8] fewer tokens per turn at statistically equal fidelity, and real-clock prefill agrees (204.7 vs. 314.1 ms, a currency carrying the jitter of Supplementary Sec. S.A(vi)). The sharpest cell is the arm that got the *smaller* budget: `comp_open12` costs 153.8 [146.7,160.7] tokens less than the view and still recomputes 19.2 [14.8,24.2] tokens more. *Pareto* (C8-P4): the view weakly dominates the three arms whose prompt is measurably *larger* than its own (`comp_open50`, `comp_open25`, `comp_keep25`) on all three coordinates. For the three arms at or below its budget the interval criterion returns no dominance in either direction, and the reason is visible in the table rather than hidden by it: `comp_keep20` is no worse on B by this test, so an order that requires "no worse on both" cannot fire even though the arm recomputes 170 tokens more per turn. No competitor arm weakly dominates the view at any rate.

b) Replication on a second architecture, with one difference worth naming: The same roster, sessions, turns, budget and nonce discipline served to `qwen2.5:14b` (3,864 calls, 0 server errors, that model's own published cost leg as the reference) reproduces the pattern. The difference is where the matched arm lands: that model's append-only view carries 459 tokens rather than 445, so its .20 arm sits just *below*

the view's budget instead of on it (ΔB -6.9 [-12.7,-1.2]); the match is architecture-specific, because a prompt's token count is a property of the tokenizer and not of the policy. The recompute gap does not move: at that arm the view recomputes 170.0 [165.8,173.8] tokens less per turn, reuse share 0.524 against the pruner's 0.907. Fidelity behaves as before: protecting the answer row keeps it (0.996 against that run's naive window at 0.987, paired $+0.009$ [-0.002,+0.020], and 0.996 against the view's 1.000), and without protection the same two rates lose the same way, because the row simply is not there any more: accuracy 0.071 at .20 and 0.000 at .12, with the gold record present in 7.1% and 0% of those contexts. The dominance verdict is identical in shape: the view weakly dominates `comp_open50`, `comp_open25` and `comp_keep25`, and returns no relation for the three arms at or below its budget, and no competitor arm dominates the view on either architecture.

c) What this licenses, and what it still leaves unmeasured: Two readings, both wanted by a system builder. Fewer tokens in the prompt is not a route to less recompute: at this budget a learned content rule can only buy compactness, and it pays for it in fidelity unless the very row the query needs is protected, and protecting it recovers the fidelity while leaving the 170-token recompute gap intact. Conversely, byte stability does not decide which record survives: our arm holds the head stable and appends, and its fidelity here is a property of what is admitted, which is the orthogonal axis of Sec. III. What remains untested is narrow, but not empty: a *generative* summarizer, which replaces text instead of deleting it, so its output is not verifiable as a subsequence and its failure mode is a rewritten value rather than an absent one; and a content rule run *inside* an append-only head, which is the combination the bounded arm of Sec. VI-B1 approaches with a deterministic fold but not with a learned selector. The ceiling of this leg is the same as the cost leg's: an in-process radix cache, single sequential requests, no concurrency and no vLLM, so this is mechanism evidence, not a production TTFT number.

F. Quality benefit on realistic retrieval

On realistic LongBench passage-retrieval danger slices, prefix-aware maintenance indicates a modest reduction in substitution relative to naive truncation, with a consistent direction across seven checkpoints of five server-reported architectures at two temperatures each. The pooling set is our pre-specified primary analysis (declared at revision) as every served (model, temperature) stratum, with no capability filter: across those 14 strata the pooled effect survives correction (Mantel–Haenszel T=18.99, p=1.3×10^{-5}; over the 7 mutually independent temperature-0 model strata T=10.88, p=9.7×10^{-4}), and the question-cluster bootstrap gives $D = 0.0782$ with 95% CI $[+0.0031, +0.1631]$ over 47 questions and 652 paired rows ($P(D \le 0) = 0.021$). The sensitivity analysis restricts by a criterion fixed without reference to the outcome (every served checkpoint of $\ge$ 7B parameters, i.e. all 14 strata minus the two `gemma2:2b` cells, 6 independent models and 12 strata) and gives T=22.15, p=2.5×10^{-6} (over the 6 independent temperature-0 models T=12.25, p=4.6×10^{-4}) and a bootstrap D=0.0860, CI $[+0.0144, +0.1679]$ ($P(D \le 0) = 0.009$, 558 paired rows). This supersedes an earlier subgroup that also filtered on the outcome (keeping only $\ge$ 7B checkpoints whose temperature-0 stratum showed $b{>}c$): that filter is visible in Table S10 and mechanically inflates the effect, so we removed it and report the size-only pool here. Both readings answer different questions: the primary all-strata set is the generalisation claim, the size-only pool is the mechanism-consistency check. Not every configuration clears a nominal threshold on its own; only `qwen2.5:14b`@t0 reaches McNemar significance alone ($p = 0.031$), and the 2B and 8B settings sit near a tie (Table S10, Fig. 5). Against the pre-declared smallest effect size of interest (Δsubst$\ge$ 0.03), the primary point estimate clears the floor but its interval does not: the lower bound sits at 0.0031, so the honest reading is *a direction-consistent benefit whose size we cannot pin down beyond* $\sim$ 0.03–0.16, not a demonstrated SESOI-sized effect. The size-only sensitivity pool does not rescue the floor either: its interval starts at 0.0144, below 0.03, so on the pre-declared criterion neither the primary nor the sensitivity reading demonstrates an SESOI-sized effect. The removal of the outcome filter costs the paper that claim, and it is the honest price. That imprecision is a property of the design, and it can be stated as a number rather than an adjective: the cluster is the question, whose 47 paired mean differences have SD 0.278, giving SE 0.041 (the normal half-width $1.96\,\mathrm{SE} = 0.079$ reproduces the bootstrap interval's 0.080). The minimum effect this design detects with 80% power is therefore $\Delta = 0.114$ (larger than the point estimate itself) and pinning the lower bound above the 0.03 floor at the observed effect would take 127 paired questions, while making 0.03 itself detectable at 80% power would take ~ 672 (`power_mde.py`, Supplementary Sec. S.B). The danger-slice benefit is thus a claim we can orient but not size, and closing that gap is a sampling problem, not an analysis choice.

We previously read the same table as a capability dependence (benefit *size* tracks capability, not architecture); tested against an external anchor that reading does not survive, and we drop it rather than re-label it. Supplementary Sec. S.G keeps the three checks in full (heterogeneity, the size-class rank correlation with its leave-one-out range, and the ordering's counter-examples); the defensible statement is *direction consistent across models, magnitude heterogeneous and undifferentiated by size*, and the near-null 2B/8B strata are reported as observations, not as evidence for a capability law. Table S11 keeps the pooling honest in both directions: dropping the two `gemma2:2b` strata lifts the Mantel–Haenszel statistic from 18.99 to 22.15 and p from 1.3×10^{-5} to 2.5×10^{-6}, so the full set is the diluted reading and we keep it as the headline. Even so, the cluster-bootstrap interval over the primary set stays clear of zero and gemma2's own direction is preserved ($b{>}c$) at a looser $\ge$ 0.55 cutoff on a larger slice (n=74). The slice is not tuned to a favourable cutoff: Table S8 reports how many of the 300 questions clear each threshold, and we analysed the $\ge$ 0.60 operating point (47 paired questions) a priori rather than by peeking at the effect; on the very-hardest

≥ 0.70 subset (n=14) the per-config signal washes out at a sample too small to power a difference.

G. Joint versus decoupled: a negative result

At the matched-token, no-eviction operating point, adding an explicit semantic value-folding lever on top of prefix-stable layout adds no additional benefit: joint versus prefix-only differs by -0.0025 in substitution (95% CI $[-0.0074, 0.0000]$) and by -0.22 tokens in recompute ($[-0.29, -0.13]$), 0.08% of the cost. Failing to clear a smallest effect size of interest would by itself leave the two arms *undetermined* rather than equal, so we tested equivalence directly: a two one-sided tests (TOST) procedure whose resample unit is the session, matching the design, against the pre-declared margins ± 0.03 substitution and ± 20 tokens. Both axes come out equivalent at α=0.05 (each one-sided bootstrap $p < 2\times10^{-4}$, and both CIs sit inside the margins), while joint versus semantic-only is equivalent on substitution ($0/0$ discordant events) and decisively *not* on cost (-262 tokens, outside the margin): the expected pattern, since only the layout lever moves cost. Under those declared margins the two levers therefore act as substitutes with a sub-additive interaction (interaction $+0.390$ base, $+0.808$ tail; Table V, Table S7). Read off the factorial decomposition in Table S7, both main effects are large and carry CIs well clear of zero (semantic dropping moves substitution by -0.392 to -0.808; the stable head halves recompute by 263–413 tokens), but the substitution *interaction* is positive with a CI excluding zero, exactly the sign of two mechanisms that remove each other's marginal value rather than compound. This holds on two model families and two answer protocols (Fig. 4). We carry this as a documented negative result, not a claim: there is no joint>decoupled effect to build a unified scorer around at this operating point.

a) What the `corr` column does and does not say: Read `corr` in Table V as a format match, not as a second fidelity readout: under the semantic lever it falls from 0.696 to 0.512 while substitution stays at 0.000, because the arm that loses it answers more often with the gold record's own text instead of its id. Supplementary Sec. S.I.00b gives the dissection (four script-computed answer categories, the paired exact test over the 408 shared turn-cells, how the categoriser's id rate differs from this table's `corr`, and the naive cell's opposite behaviour). The fidelity claims in this paper ride on substitution and on the forced-commit protocol of Fig. 4, where the response space is a single value and format has nowhere to go.

H. Reading the axes against the objective

Recall the constrained objective of Eq. (2), read with the dominance order of Sec. III: $\pi \succ \pi'$ iff π is no worse on both B and R and strictly better on one, among policies with $F \geq \tau$. The cost results above place each arm in the (B, R) plane at the tested operating point, and the external compressor of Sec. VI-B2 is the only point in that plane that is not one of our own policies. The naive window is outside Π_τ and outside $\mathcal{C}$: its front shifts every turn, so $\ell_t \approx 0$, it pays $\Theta(B)$ recompute, and it fails F in near-twin-rich regimes (substitution up to 0.808). The stable head enters $\mathcal{C}$ and, at *matched* F, is better than the naive window on both coordinates (B 445 vs. 1575 tokens carried, R 249 vs. 1557 recomputed), so it *dominates* it in the sense of the order above rather than by any weighting choice. The pinned control is also dominated on both coordinates (B 1584 vs. 445, R 285 vs. 249), but it was never a frontier candidate: it exists to hold B fixed and vary only whether the front is stable, so as an instrument it succeeds precisely by being a bad policy: freezing the head is what a view that must reflect a changing relation cannot do. The compressor lands where the formulation says an orthogonal axis should land: pruned to the view's own budget it recomputes 170.1 [165.9,173.8] tokens more per turn at statistically equal B and equal F, so it is not weakly dominated by compactness and does not dominate the view in return; the two points are separated by the stability of the front, not by how much text is sent. M-A is thus the measured point that improves R *and* B together while holding F, and the layout bonus (suppressing substitution at d=0) is why the semantic admission lever, which targets F alone, has nothing left to contribute here. Two limits on this reading are worth stating: dominance is established for the arms we measured, not proved over $\mathcal{C}$, and the stable-versus-pinned R gap (36 tokens) clears the practical floor of Sec. VI-A while the amortized-cost argument for preferring M-A over pinning rests on B, which no fidelity floor constrains. The frontier at a bounded budget is only partly mapped (T1).

I. Float material held here in full

The symbol table of Sec. III and the axis-by-axis comparison of prior work (Sec. II) are given in full here, with the caption clauses the body floats do not print and the body prose of Sec. S.J.10. The measured numbers stay in the run files and in the body tables.

a) Caption note: the prior-work table (Table S17):

b) Caption note: Algorithm 1:

c) Caption note: the schematic figure (Figure 1):

d) Caption note: the cost table (Table II):

e) Caption note: the competitor table (Table III):

f) Caption note: the 2×2 fidelity table (Table V):

J. Body prose given in full

Each block below is the full text of the body item named in its heading, moved verbatim and kept in body order. Its heading gives that item—the body section, or the table or figure whose caption the body shortened—followed by what the block supplies. The body keeps the finding and the statistic that carries it for each measured leg, and no measured number left the manuscript.

a) Figure 1 caption in full — the one isolated variable: The one isolated variable: where older history sits relative to the radix prefix cache. A recency window (A) rewrites the leading bytes every turn, so the whole prompt is re-prefilled; M-A (B) keeps an append-only head whose leading bytes never change and confines churn to a small volatile tail. Gold

values are present verbatim in both arms; the arms differ only in prefix stability. (C) is the view the position stress of Sec. VI-C motivates and Sec. VI-D tests: a bounded block of current bindings, inserted between head and tail and *expected* to be rewritten, so invalidation is confined to block plus tail and sized by the live-binding count I.

b) Body Sec. VI-A in full — run inventory, model roster and call budget: Supplementary Table S4 (Sec. S.E) traces each multi-turn study from generated records to analysed cells, per arm, computed from the run files (`sample_flow.py`). The analysis window is a *turn range fixed by the generator before any model call* and the same rule for every arm of a study, so designed cells equal calls issued equal cells analysed and there is no post-hoc exclusion. The confirmatory analyses rest on 34,134 calls in 38 named run files, all counted; one further file (276 calls) is the latency-replication diagnostic of Supplementary Sec. S.A, kept in the audit so no call goes uncounted (34,410 calls in 39 files in all) but carrying no inferential claim. Four legs are counted separately (rewritable-index fidelity and cost, the external content rule of Sec. VI-B2, and the shared-key re-servation behind Supplementary Table S20); their call counts and arm rosters are accounted in Sec. S.E. All models are released checkpoints of publicly documented families [3], [4], [5], [6], [7] named by the server's own tags: two 14B architectures for the cost harness, seven checkpoints from 2B to 27B spanning five distinct server-reported architectures at two temperatures for the fidelity passage sweep, one architecture for the 2×2 and layout factorial, and a second with a replicate epoch for the forced-commit tail stress. Supplementary Table S1 lists each model's exact per-experiment call count over the 38 confirmatory files, records how the server's tags collapse and names the 39th file as the check vi replication.

c) Body Sec. VI-A in full — determinism, nonces and the shipped package: Each experiment is a deterministic replay of a generated session set. Randomness is fixed by seed (`1234/777`) and temperature (0 and 0.5–0.7); *per-config and per-run nonces* are injected into each arm's prompt so a stale whole-prompt radix node from a prior invocation can never be credited as a cache hit. Record content is always real LongBench text [8] (two materials, Supplementary Sec. S.D). The fidelity floor (gold present in every arm and turn) is asserted before any substitution claim, and `correct` is read only under the forced single-value commit protocol. Supplementary Table S2 names the script triple behind every number and Table S3 collects every run on one axis set. `headline_numbers.py` re-reads the per-prompt event logs and recomputes every mean, ratio, risk difference, CI and p-value quoted here, so a transcription slip shows up as a disagreement with that output. The package (code, gates, per-call logs and regenerators) ships with this submission as an anonymised artefact.

d) Table II caption in full — three arms at matched accuracy: **Cost axis (C2): three arms at matched answer accuracy** (`qwen3:14b`, temp 0, 552 paired turns / 24 sessions; `qwen2.5:14b` reproduces the ordering; — = not tabulated for the replication leg). Recompute = prompt_eval_count − prompt_eval_cached_count (tokens), prefill = prompt_eval_duration (ms). Fidelity is F_{acc} (exact-match), not F_{subst}: no competing value is in context here. Run files: Supplementary Table S2, row "Cost, 3 arms".

e) Figure 3 caption in full — recompute at matched fidelity: Recompute cost at matched answer-fidelity (accuracy 1.000 in all three arms), qwen3:14b, temp 0, 552 paired turns / 24 sessions (bars = mean recompute tokens/turn = `prompt_eval_count` − `prompt_eval_cached_count`; prefill ms beneath). M-A's prefix-stable head recomputes $\sim$ 6.26× fewer tokens than the naive sliding window (session-cluster bootstrap D=1308.6, CI $[1280.1, 1331.0]$, $P(\leq 0)$=0). The `pinned_raw` control holds prompt *size* constant and reproduces 97% of the gap, attributing the win to prefix invalidation rather than fewer tokens sent; `qwen2.5:14b` reproduces 6.4×.

f) Body Sec. VI-B2 in full — pruner arm design and controls: Token-level compression buys compactness, not reuse. A LLMLingua-2 token classifier (`llmlingua-2-bert-base-multilingual-cased-meetingbank`, 177M parameters, CPU, serving model untouched) scores the whitespace tokens of the naive window and keeps the top fraction [9]. Seven arms run over the *same* 552 (session, turn) cells, 24 sessions, turns 7–29, with the same pinned `num_ctx`, seed, temperature and per-arm ascending-turn cache streams as the cost leg, `length_only` replayed as a same-run control: 3,864 calls per architecture, 0 server errors.

Keep-rates .50/.25/.20/.12 are *open* (answer row unprotected) and .25/.20 re-run *protected* (`comp_keep`*, the row force-kept, benefit of the doubt); binding survival is counted by string test in every prompt, never assumed. `check_comp_controls` (all 552 trials) asserts each compressed prompt is a whitespace-token *subsequence* of the same trial's naive window and prints binding survival (551/552, 338/552, 39/552, 0/552 at the four open rates; 552/552 protected). Offline, before any call, a competitor prompt preserves $\approx$ 0.00 of its own previous turn's leading bytes against 0.242 for `prefix_stable` (the consecutive-pair mean over turns 7–29 of *this* stream, a different window from the 0.640 mature-head figure of the layout gate of Sec. IV).

Four predictions were fixed beforehand (**C8-P1** budget, **C8-P2** protected-arm fidelity, **C8-P3** recompute, **C8-P4** non-domination of the view; stated in full in Supplementary Sec. S.O). Two protocol amendments were made during this leg: absolute recompute against `naive` is confounded by B, so the leg reads on R/B and paired differences; and the matched keep-rate .20 was chosen by a fit over partially served arms, whose full-leg resolution names an arm (.21) never served. Both are disclosed with rationale in Supplementary Sec. S.O.

g) Body Sec. VI-B2 in full — result on the three coordinates: Table III gives the three coordinates. *Budget* (C8-P1): three arms sit at or below the view's

445.5 tokens (`comp_open20` 434.2, `comp_open12` 291.7, `comp_keep20` 442.6), so the comparison is size-matched where it needs to be.

Fidelity (C8-P2): at the view's own budget the unprotected pruner keeps the binding in 39 of 552 contexts and is correct in those same 39 (0.071 vs. the naive window's 1.000; paired -0.929 [-0.947,-0.909], $p = 7 \times 10^{-155}$ over 513 discordant pairs, none the other way). Attribution is exact cell by cell: the correctly answered cells *are* those where the binding survived, so the loss is the deleted row rather than attention to it. The protected arm confirms it from the other side (row survives in all 552, correctness 0.998 vs. 1.000, paired -0.002 [-0.005,0.000], one discordant cell, inside the floor).

Recompute (C8-P3): as pre-declared every pruned arm recomputes fewer absolute tokens than the naive window (-597.1 to $-1{,}289.4$, all intervals below zero), which is Amendment 1's confound rather than the mechanism. The mechanism is the share R/B (from unrounded means): 0.919–0.972 for every compressed arm, 0.989 naive, 0.558 for the view. Pruning to a fifth of the words buys a smaller prompt that is still almost entirely recomputed. At the matched cell (`comp_keep20`, $\Delta B = -2.9$ [-8.7,+2.8]) the view recomputes 170.1 [165.9,173.8] fewer tokens/turn at statistically equal fidelity, and real-clock prefill agrees (204.7 vs. 314.1 ms). The sharpest cell got the *smaller* budget: `comp_open12` costs 153.8 [146.7,160.7] tokens less than the view yet still recomputes 19.2 [14.8,24.2] more.

Pareto (C8-P4): the view weakly dominates the three arms whose prompt is measurably *larger* (`comp_open50`, `comp_open25`, `comp_keep25`) on all three coordinates, and for the three at or below its budget the interval criterion returns no dominance either way. `comp_keep20` is no worse on B, so an order requiring "no worse on both" cannot fire though it recomputes 170 tokens more per turn. No competitor arm dominates the view at any rate.

h) Replication on a second architecture: The same roster, sessions, turns, budget and nonce discipline on `qwen2.5:14b` (3,864 calls, 0 errors, its own cost leg as reference) reproduces the pattern, with the match itself moving: that view carries 459 tokens rather than 445, so its .20 arm lands *below* the budget instead of on it (ΔB -6.9 [-12.7,-1.2]): token counts belong to the tokenizer, not the policy. The gap does not move (170.0 [165.8,173.8] tokens/turn less, reuse share 0.524 vs. 0.907), protection still keeps fidelity (0.996 vs. naive 0.987 there, paired $+0.009$ [-0.002,+0.020]; 0.996 vs. the view's 1.000), and unprotected the same two rates lose because the row is gone (0.071 at .20, 0.000 at .12; gold present in 7.1% and 0% of contexts). No arm dominates the view on either architecture. What this licenses: fewer tokens is not a route to less recompute, and byte stability does not decide which record survives (orthogonal axes, Sec. III). This leg runs on the cost leg's serving setup (in-process radix cache, sequential requests, no vLLM) and leaves a *generative* summarizer and a content rule run *inside* an append-only head open (Supplementary Sec. S.J).

i) Body Sec. VI-C in full — stability, not labelling: Stability, not labelling. Crossing the two things the stable arm changes at once (byte-fixing order and block labels) on identical content (d=0, gold and near-twins verbatim; $12 \times 34 \times 4 = 1632$ calls, 0 server errors; Table S14, Supplementary Sec. S.H) separates them without a model call: front reuse is 2.0%/7.1% in the sliding cells against 51.2%/51.3% in the stable ones, competing values present are 1.06 in all four cells, and total tokens spread by 1.97% (the $\approx$ 12-token header asymmetry, reported not hidden). Stripping labels from the stable arm leaves substitution at $0/408$ (Wilson upper 0.009); adding them to the sliding arm moves $0.473 \rightarrow 0.387$ (exact McNemar b=59, c=94, $p = 0.006$), so labelling helps only where the front already slides. Cost responds the same way: label contrast $+6.1$ tokens/turn against stability contrasts -260.4 $[-273.0, -247.7]$ and -268.9 $[-282.2, -255.6]$; the session-clustered linear-probability refit attributes -0.473 ($p = 1.3\times10^{-8}$) to stability and -0.086 ($p = 0.081$) to labels, and only stability clears the floor (Supplementary Sec. S.J). What survives is a *placement* claim, not a wording claim: the accumulated record is an older-first sequence whose leading bytes never move while the freshly-slid competitor sits in a separate volatile region. Two loose ends, both reported: correct answers are *lower* in the labelled stable cell (0.608 vs. 0.828), entirely in a third bucket naming neither record (0.392 vs. 0.172), so labels are not style-neutral; and a second-epoch re-run reproduces only 56% of that cell's 408 answer strings byte-identically while substitution agrees on $407/408$, so cost claims ride on the recompute counters and fidelity claims on substitution, which replicated.

j) Table IV caption in full — position stress inside the head: **Position stress inside the append-only head** (`check_position_controls` passed before serving; permutation only, identical bytes and line multiset across slots). H is the number of head lines at the probed turn; one competing near-twin value is present in the head in every cell, so the hazard is fixed and only the gold line's position moves. 120 sessions per cell per model, temp 0. Prompt size is *not* a manipulated axis and is reported here rather than as a column because it is a mean over the cell's 120 calls, and the two tokenizers differ: `qwen3:14b` averages 529 tokens at H=6 (max 713, min 400) and 1271 at H=24 (max 1650, min 1007); `llama3:latest` averages 494 (max 677) and 1191 (max 1567). The maxima are the 1650/1567 quoted in Table S4, i.e. 20.1%/19.1% of `num_ctx`.

k) Body Sec. VI-C in full — whether the head itself is immune: No. The factorial probes the *newest* record in the tail, leaving open where a line sits *inside* the head, the condition lost-in-the-middle [10], [11] predicts to be worst. Permuting the gold line within an otherwise fixed head (120 sessions $\times$ $H \in \{6, 24\}$ $\times$ three slots, one competing near-twin verbatim in every cell, 720 calls per model on two architectures, 0 server errors; the offline gate confirms the permutation changed nothing else) diagnoses where the model reads, not a policy arm: in an append-only head, moving a line forward is the edit that forfeits the prefix.

Placement costs $30\times$ the declared floor. With gold at the head's front, substitution is 0.000 (Wilson upper 0.031) at both lengths and both models; moving the same line to the middle gives $0.550/0.742$ (H=6/H=24, `qwen3:14b`) and $0.158/0.325$ (`llama3:latest`), with correctness collapsing $0.942/0.933 \rightarrow 0.042/0.008$ and $0.375/0.475 \rightarrow 0.008/0.000$; middle−first at H=24 is -0.925 ($[-0.967, -0.875]$, $0/111$ discordant, $p = 8\times10^{-34}$) and -0.475 ($[-0.567, -0.383]$, $0/57$, $p = 1\times10^{-17}$), the largest fidelity effects measured here, same-signed on both architectures. The gradient is front-dominant and *monotone*, not U-shaped (middle vs. back -0.108, p=0.047): primacy, not recency recovery. Length costs nothing at the front (correct -0.008, p=1) and deepens the deficit elsewhere (substitution $+0.192$, p=0.001; clustered linear probability model (LPM) middle×length p=0.0007). This narrows our own claim: the stable arm's 0.000 is *region separation*, not immunity to position. Inside the head the front line confers authority on the *oldest* surviving record, safe under our semantics only because the *current* binding is what a query asks for and lives in the tail. A workload recalling a historical binding faces that gradient with no cheap handle inside one append-only view, which motivates the second view and its price (Sec. VI-D); the per-slot grid (Table IV), conversions and remaining contrasts are Supplementary Sec. S.N.

l) Table V caption in full — joint versus decoupled: **Joint-vs-decoupled 2×2 (C5), `qwen3:14b`, temp 0, 408 turn-cells per config** (run files: Supplementary Table S2, row "Joint 2×2"). `c00` naive (d=0, p=0), `c10` semantic $(1, 0)$, `c01` prefix $(0, 1)$, `c11` joint $(1, 1)$: d folds competing near-twins, p puts the stable head behind the sliding one. Blocks are the base layout (gold deep) and the tail-distractor stress (both near-twins in the volatile tail). Only *subst* is a fidelity readout; `corr` is the strict verbatim record-id match (Sec. VI-C) and `other` names neither gold nor competitor (column partition and rounding note: Supplementary Sec. S.J.9).

m) Interaction plot caption in full (plot omitted from the body for space): Substitution under the 2×2 factorial (semantic-distraction d × prefix-depth p), qwen3:14b, no-eviction operating point. The naive arm (d=0, p=0) collapses (0.392 with a deep distractor, rising to 0.808 when a near-twin is parked in the volatile tail), while prefix-stability alone (d=0, p=1) already drives substitution to $0.002/0.000$. The d=1 line is flat at 0: adding the semantic lever on top of a stable head buys nothing. The lines converge rather than cross favourably, i.e. the interaction is *sub-additive* ($+0.390/+0.808$): the two levers are *substitutes*, not complements. This is the decisive result against the joint admission hypothesis, carried as a negative finding (Sec. VI-B1).

n) Figure 4 caption in full — forced-commit tail stress: **Generality of the 2×2 (C5), forced-commit protocol, tail stress, over three runs at n=120 cells each** (12 sessions × 10 mature turns; 0 server errors). Two *independent* epochs on `qwen3:14b` and one on a second family, `llama3:latest`; the naive arm's substitution spread (0.550–0.967) is reported as a spread, not a single-family headline, and the second family is *more* fragile, not less (b=116/c=0 McNemar against the stable arm, all three runs $p < 10^{-19}$). Substitution is 0.000 in the prefix-only `c01` and joint `c11` cells of every run. Recompute *is not* family-invariant in absolute value (the `llama3` tokenizer reaches a different byte count) but the mechanism is: prefix stability roughly halves the naive recompute in both ($810.7 \rightarrow 397.8$ `qwen3`, $765.6 \rightarrow 370.7$ `llama3`), and the joint arm's edge over prefix-only is -1.0 token (`qwen3`) and $+1.6$ tokens (`llama3`), both far below the 20-token actionable floor of Sec. VI-A, so the semantic lever adds nothing measurable in either family. Run files: Supplementary Table S2, rows "Tail stress (forced)" and its second architecture. Each column is read from the run file whose rows record the tag it is labelled with: two `qwen3:14b` passes for the first two columns and the `llama3:latest` run for the third.

o) Body Sec. VI-D in full — why a rewritable view, and its design: If the only handle on a measured fidelity failure forfeits the cached prefix, the maintenance model is missing a view; Sec. V-B argued the missing view mechanically and we adopted and tested it rather than cite it. `plain` re-runs the published stress under re-keying (`RELATED`k `->` W), since an index is only well defined over distinct keys; `index` renders `STATE` ‖ `INDEX` ‖ `RECENT` with one `key -> value` line per live binding including the queried key; `index_nok` is a placebo carrying the same block in the same position with the queried row removed (one line shorter, a 15-character difference), so presence, framing and query distance are fixed and the answer row is the only content difference. Same 120 sessions, head lengths and slots as the stress: 720 calls per arm per model over two architectures, 4,320 from one 2,160-trial grid. `check_index_controls` passes on all trials before any call; the block's character cost and the verbatim gate list are Supplementary Sec. S.N. Pre-declared: **P1** gradient present in `plain`, **P2** it shrinks inside the 0.03 floor in `index`, **P3** the fix rides on the answer row.

p) Body Sec. VI-D in full — fidelity result: Supplementary Tables S19 and S21 (Sec. S.N): no server errors, no exclusion, 120 sessions per cell, the two architectures read differently and are reported separately. *P1 holds on qwen3 but is not met on llama3*: re-keyed `plain` shows the gradient at every contrast on qwen3 (middle−first -0.217 [-0.292,-0.142] at H=6, -0.375 [-0.467,-0.283] at H=24), while llama3 sits between 0.967 and 1.000 in five of six cells, so its `index` numbers are a ceiling comparison. *P2 is not met as written: the block attenuates the gradient, it does not flatten it.* Within `index`, middle−first is -0.050 [-0.117,+0.017] at H=6 (inside the floor) but -0.158 [-0.233,-0.083], $p = 6.6\times10^{-5}$ at H=24 (outside): the block *removes* 77% of the short-head and 58% of the long-head gradient, and the residual is why P2 is not met. The difference-in-differences says the block moves the gradient rather than the level ($+0.167$ [+0.092,+0.250], $+0.217$ [+0.133,+0.300]; on llama3 `index` is 1.000 in all six cells, so P2 is met there only because P1 is not). *P3 holds where there is a gradient*: index−placebo at disadvantaged slots is $+0.117$ to $+0.400$ (all $p \leq 0.0013$; per-slot intervals in Supplementary Sec. S.J) while the placebo keeps its gradient: the gain rides on the

answer row, not on a second block near the query.

What it does not buy: at the front slot, where arrival already put the answer, `index` is not better: -0.033 [-0.067,-0.008], a loss at the pre-declared 0.03 magnitude which four discordant pairs cannot resolve (exact p=0.125). It redistributes correctness instead of raising the level. Substitution is 0.000 in every cell of both architectures against 0.550–0.742 in the shared-key stress. That is no re-keying artifact: served on the shared-key `STATE`, a maintained `INDEX` still recovers the collapse (index−plain correct $+0.479$/$+0.662$) while a stale index drives substitution above the raw state (Supplementary Table S20). It locates C7: the earlier collapse was shared-key hazard and position together, the position effect alone being the -0.217/-0.375 gradient. A post-hoc split by answer form (Supplementary Table S16) puts the residual in `plain` as prose restatement of the gold record without its identifier (0.237 against 0.069 in `index`): the block repairs the report of the identifier, not retrieval of the row.

q) Body Sec. VI-D in full — what the block costs: Its price is linear in the number of bindings named, and on this workload it is not cheaper than the fold. Measured on *published* sessions regenerated from the frozen seed and byte-checked against the published trial file before any call: 20 of the cost leg's 24 sessions (460 of 552 turns), the width the leg's generator records as its default, fixed before any call, not an exclusion, hence reference means of 453/250.8/1,566.8 against Table II's 445.5/248.7/1,557.3, every one of the 460 prompts byte-identical. The premium over `prefix_stable` grows with block width: $+27.7$ [27.1,28.2], $+91.8$ [89.5,93.9], $+195.7$ [190.3,200.7] tokens/turn at I=2, 8, 24, correctness 1.000 in all five arms over the same 460 paired turns, so the matched-quality premise holds unadjusted. Regressing the paired premium on I: 7.40 tokens per `INDEX` row (standard error (SE) 0.11, $p = 3.6\times10^{-24}$), intercept $+21.1$ (the header line), $1{,}380$ observations clustered on 20 sessions, the fit within 11.4 tokens of every level. Three levels of I fix two parameters, so the residual degrees of freedom are zero and the marginal price (10.7 then 6.5 tokens per added row) leaves room for saturation: we claim monotone growth, compatible with $\Theta(I)$, not a tested form. The history-length flatness is the separate Sec. V-B leg, not jointly varied here. Because every turn mints a fresh key ($\texttt{OBS}_t$) and nothing is superseded, that dependence on I is itself the evidence that the block is re-prefilled from its first byte each turn: 7.40 is the cost of *re-rendering*, an upper bound on the incremental single-row update a supersession would need (T7). In absolute terms I=24 takes the prompt $453 \to 650$ tokens and recompute $250.8 \to 446.6$ ($\times1.78$) while staying $3.51\times$ below the naive window's $1{,}566.8$; hit rate falls $44.6\% \to 31.3\%$ and prefill rises $+28.0$ to $+117.4$ ms, a jittered currency (Supplementary Sec. S.A(vi)) which is why the claim is in tokens. A bounded head folds once per m=6 appends ($+209$ recomputed tokens on the turn it fires, 25.3/turn amortized, 23.8 net vs the unbounded arm) while a rewritable index pays $\Theta(I)$ *every* turn: at I=24 the second view costs about a worst-case fold, always, and at I=2 the premium (27.7) matches the fold's surcharge, so a narrow index is affordable *alongside* a bounded head, returning the $+0.117$ to $+0.400$ exact-match correctness of Sec. VI-D (not fidelity, substitution being 0.000 in that grid) that no fold setting buys.

r) Figure 5 caption in full — per-config effect on the danger slice: Per-config effect of M-A over naive truncation on the harder *passage* danger slice (LongBench-style retrieval where the naive window retains a near-adversary passage, `lo_max`$\geq$0.6; $n\approx47$ paired questions per stratum). Each dot is Δ substitution = length_only − M-A = $(b-c)/n$ for one (model, temperature, seed) config; positive favours M-A. M-A is strictly below naive in 12/14 strata and exactly ties the two `qwen3:8b` strata. Whiskers are 95% normal-approximation intervals on the matched difference, $\Delta \pm 1.96\sqrt{b+c}/n$ (n=44 for the two `qwen3.8:27b` strata), tabulated in Table S10 and drawn to scale the noise within a stratum; they are not the test statistic, which is exact McNemar on the same discordant counts (Table S10). The axis is deliberately not truncated at zero: 12 of the 14 stratum intervals cross it, only `qwen2.5:14b`@t0 is individually significant ($\star$, p=0.031), and `mistral` t=0.6 is not starred although its interval has a positive lower bound, because its exact p is 0.065. The dotted line is the pre-declared smallest effect of interest ($\Delta\geq$0.03, Sec. VI-A). Below the rule, two diamonds give the question-cluster bootstrap pools: filled, the primary pool over *all* 14 strata (no capability filter, pre-specified at revision) at D=0.0782, CI $[+0.003, +0.163]$, $P(\leq 0)$=0.021; open, the size-only $\geq$ 7B sensitivity pool (12 strata) at D=0.0860, CI $[+0.014, +0.168]$. Pooling reaches an interval no single stratum can, yet its lower end still sits below the floor, which is the reading the body reports: a *direction-consistent* benefit with *magnitude the study cannot resolve*. Per-model sizes range over 0.00–0.149 but are not distinguishable from sampling noise (Cochran Q, p=0.48) and do not track checkpoint scale (Table S12). Reported as such, not as a headline per-config result.

s) Body Sec. VI-C5 in full — quality benefit over seven checkpoints: On realistic LongBench passage-retrieval danger slices, seven checkpoints of five server-reported architectures at two temperatures each favour prefix-aware maintenance over naive truncation. The primary pooling set is pre-specified (declared at revision): every served (model, temperature) stratum, no capability filter. Across those 14 strata the pooled effect survives correction (Mantel–Haenszel T=18.99, p=1.3×10^{-5}; over the 7 mutually independent temperature-0 strata T=10.88, p=9.7×10^{-4}) and the question-cluster bootstrap gives $D = 0.0782$, CI $[+0.0031, +0.1631]$ over 47 questions and 652 paired rows ($P(D \leq 0) = 0.021$). A size-only sensitivity pool ($\geq$ 7B, 6 models, 12 strata), restricted by a criterion fixed without reference to the outcome, sharpens the statistic but not the interval; the outcome-filtered subgroup it supersedes inflated the effect and was removed, and all three readings are unabridged in Supplementary Sec. S.B. Only `qwen2.5:14b`@t0 reaches McNemar significance alone ($p = 0.031$) and the 2B/8B settings sit near a tie (Table S10, Fig. 5). Against the SESOI (Δsubst≥ 0.03) the point estimate

clears the floor but neither interval does (lower bounds 0.0031 primary, 0.0144 size-only), so the honest reading is a benefit whose size we cannot pin beyond ~ 0.03–0.16. The imprecision is a design property with a number: with the question as cluster the 80%-power minimum detectable effect (MDE) exceeds the point estimate and clearing the floor needs roughly three times the paired questions (`power_mde.py`, Supplementary Sec. S.B), a sampling problem, not an analysis choice.

The defensible statement is *direction consistent across models, magnitude heterogeneous and undifferentiated by size*, with near-null 2B/8B strata as observations, not a capability law; an earlier capability-dependence reading does not survive an external anchor and is dropped (Supplementary Sec. S.G: heterogeneity test, size-class rank correlation, counter-examples). Table S11 keeps the pooling honest: the full 14-stratum diluted set stays the headline with its primary interval clear of zero, and gemma2's direction survives a looser cutoff on a larger slice (drop-one statistic, ≥ 0.55/n=74 and ≥ 0.70/n=14 readings in Supplementary Sec. S.B). The slice is not tuned to a favourable cutoff: Table S8 gives how many of the 300 questions clear each threshold; we analysed ≥ 0.60 (47 paired questions) a priori.

S.K. Design detail: the deterministic gate and the boundary of the index semantics

Two passages of Sec. IV are given here in full: the offline layout gate that separates the manipulated variable before any model call, and the explicit statement of what the rewritable-index rendering defines versus leaves open. Nothing here is new data; both are the body text in full.

a) The deterministic gate, in full: Because the whole point is to isolate *layout* as the manipulated variable, we gate each generated session set with a deterministic checker (`check_joint_layout.py`) that runs *before* the first inference call and fails the run if the axes are not actually separated. On the base 2×2 it confirms (A) competing values present 1.06 under d=0 versus 0.00 under d=1; (B) leading-byte stable-prefix fraction 0.018 for the sliding layout versus 0.640 for the mature stable head; and (C) prompt size matched across the p axis to within 1.4%. On the tail-stress rework the same gate re-passes with competing values 1.60 versus 0.00 and prefix fraction 0.011 versus 0.653, i.e. the substitution hazard is genuinely live in the prefix-only arm even though its head is stable. This is what lets us attribute a downstream effect to the mechanism rather than to an accidental prompt-size or content difference between arms.

b) What the paper defines and what it leaves open, in full: What this paper defines and what it leaves open is stated plainly here, because the distinction is the point of the measurement, not a caveat. The rendering rule is defined and exercised: one row per key, rows re-emitted in place, block bytes free to change while the head's stay fixed. The *maintenance algebra* is not. Neither leg produces a workload in which a key already in the block is re-bound: the fidelity grid has two value-bearing keys per session, the cost stream gives every turn a fresh key ($\texttt{OBS}_t$), so supersession is never measured, and in the cost leg the block is re-rendered in full at every turn, which prices a *whole-block rewrite* rather than a single-row update. The two rules the algebra would need are therefore specified as choices, not tested: which rows are live (the paper's arms use arrival-recency, I most recent bindings; a real controller could use reference reachability or a session-scoped TTL, and neither is measured here), and what happens when live keys exceed $I_{\max}$ (evicting a row is a utility decision whose price this study does not estimate). The same gap covers head-versus-index conflicts: if a key appears in `STATE` with a different value than in `INDEX`, the arms of this paper never present that case, so which record a query reads is undefined by the experiment. Sec. VII records this as a scope threat and Sec. VI-D reports only what the rendered block buys and costs.

S.L. Problem-derivation detail

The two readings that Sec. III states briefly, written out here in full: the cache-hit reading of the ℓ_t term, and the fold-period correction behind Eq. (3). Every number below also appears in the body.

a) Cache-hit reading: The measured cache-hit rates make the ℓ_t term directly visible: the naive window hits the reusable prefix on only 1.1% of bytes (its front shifts, so $\ell_t \approx 0$ and almost every token is re-prefilled), the stable head hits 44.2%, and the frozen-pin control hits 82.0%. Since recompute is $|P_t|$ times the *miss* rate, the stable head's larger miss rate (55.8% vs. pinned's 18.0%) still yields smaller absolute recompute (249 vs. 285 tokens) because it carries fewer than a third as many tokens. This is exactly the (B, R) vector of Eq. (2) refusing to collapse to a scalar: stability and compactness move R through different terms, and only the stable head moves both at once.

b) Fold period and realized frequency: The head is capped at $H_{\max}$ digest lines and, on overflow, the oldest $H_{\max} - F$ lines are folded into one summary line (keeping the F newest), which bounds memory at $H_t \leq (H_{\max} + 1)\Delta$ for all t and turns the flat recompute into a sawtooth. Its steady-state period is $m = H_{\max} - F$, not $H_{\max} - F + 1$: the summary line occupies one of the F kept slots, so a fold leaves $F + 1$ lines and the next overflow comes m appends later (6 turns at $H_{\max} = 10$, $F = 4$, which is what the recorded inter-fold gaps show). Over an analysis window of T consecutive turns the *realized* frequency is lower because the head must first grow to the cap: with first fold at t_1, $\bar{f} = \max\big(0, \lfloor (T - t_1)/m \rfloor + 1\big)/T$.

c) Two separations keep the framing honest: Definitional vs. measured: because $\mathcal{C}$ is defined by the append-only head, (i) *by construction* every $\pi \in \mathcal{C}$ has head bytes appended only between folds, so ℓ_t never recedes and $\bar{R}(\pi) = \Theta(V + \bar{f}\, H_{\max}\Delta)$ (Sec. V-B), so no measurement can change that; (ii) *empirically*, joining $\mathcal{C}$ costs no fidelity (at matched quality the stable head's substitution is no worse than the naive window's, and lower in 12 of the 14 served (model, temperature) strata); (iii) *empirically*, the naive window is not in $\mathcal{C}$ ($\ell_t \approx 0$) and loses on both axes at once, paying $\Theta(B)$ recompute while still failing F where near-twins accumulate. Only (ii) and (iii) are results; (i) is why they are worth measuring. *Why the*

constraint binds: any edit that rewrites a leading byte resets ℓ_t to the last stable position, and summarization and truncation both act there, so the operations that naively reduce B are the ones that destroy ℓ_t and inflate R. That is the precise sense in which the three objectives are not separable, and it motivates folding old content into an append-only head rather than editing the front.

d) Three rules make the block a view rather than a second copy: These are what the tested arms implement: *liveness* (a key is live at t iff its latest binding is present in the view the render reads, so the index is a projection over records the head has not dropped, not an independent store); *refresh* (each append enters the block, the oldest row leaves past $I_{\max}$, rows emit in arrival order as one `key -> value` line each, and only the block's bytes change, which confines the invalidation to $\Theta(I_{\max}\Delta_{\text{idx}})$); and *no conflict to resolve* (the block derives from the rows the head holds, so head and index cannot disagree, and re-emission every turn is the cost of that guarantee). Two alternatives are *not* measured: a TTL or reference-reachability rule for retiring a key, and any policy for choosing *which* bindings deserve a row once the live set exceeds $I_{\max}$ (never binding at the $I_{\max} \geq 2$ of our probes). Both are open (Sec. VII).

e) The reason to prefer the framing over the fold: It is not only the amortized term. Moving one line to a different position *inside* an append-only head re-emits every byte that follows it, the $\Theta(H)$ whole-prefix cost of relocating a single line. If placement inside the head is expensive to change, the position gradient measured in Sec. VI-C has no cost-cheap remedy; if current bindings live in a block that is rewritten anyway, "where in the head does the record sit?" loses its object for queries asking the *current* value. That turns our own narrowing into a falsifiable prediction, stated before running it: a rewritable index should flatten the head-internal gradient, and should do so because the index carries the answer row rather than because a second block sits near the end of the prompt. Sec. VI-D tests both halves with a placebo arm that keeps the block and drops the answer row.

S.M. Bounded-head arm: the fold-cost accounting

Full accounting for the finite-budget arm summarized in Sec. VI-B1, including the controls the body reports only as headline differences.

a) Does the win survive a finite head budget?: The headline arm lets the append-only head grow, which reads as an idealization: a deployed agent must fold within the head or hit the KV budget B_{KV} at $t^* = (B_{\text{KV}} - H_0)/\Delta$. We therefore re-run the same matched-quality protocol with a *budgeted* arm that caps the head at $H_{\max} = 10$ digest lines and folds the oldest into a summary on overflow (keep $F = 4$), over 12 sessions and 396 turns (48 fold events). The sawtooth of Eq. (3) appears in the raw counters exactly as predicted: append turns recompute 246.7 tokens (the $\Theta(V)$ floor, matching the unbounded arm's 248.7), fold turns spike to 456.1 (the $\Theta(H_{\max}\Delta)$ term when the head's leading bytes are rewritten). Two corrections to the closed-form account come out of the measurement rather than the model. First the steady-state cycle is $H_{\max} - F = 6$ turns, not $H_{\max} - F + 1 = 7$: the rolled-up summary line consumes one of the F kept slots, so a fold leaves the head at 5 lines rather than 4, and the recorded inter-fold gaps are $6, 6, 6$ in every session. Second, the fold *schedule* is predicted a priori, not read off the mean. The analysis window covers turns 7–39, i.e. $T = 33$ analyzed turns per session; starting from the cap-filling first fold at turn 17 with the steady-state period $m = 6$, the closed form places folds at turns $17, 23, 29, 35$ (4 per session), and the recorded inter-fold gaps are exactly $6, 6, 6$ in all 12 sessions, a falsifiable prediction that holds. With that fold count, the amortized mean works out to $246.7 + (4/33)(456.1 - 246.7) = 272.1$, which reproduces the measured 272.1 tokens/turn; because it recombines the *measured* append and fold means by the *measured* fold rate, this is a consistency check on the accounting rather than an independent prediction, and we label it as such. The bounded head stays *flat* in size (mean 454 tokens) where the unbounded head already carries 498 at these turns and keeps growing. The prefix-cache advantage is essentially budget-invariant, and we state it in one currency (absolute tokens) rather than mixing ratios and percentages: naive truncation recomputes 1564.7 tokens/turn against the budgeted arm's 272.1, a saving of 1292.6 tokens/turn (CI $[1271.1, 1315.5]$, $p < 0.01$); the same comparison on the unbounded arm of *this* run saves 1316.4 ($1564.7 - 248.3$), so enforcing a finite KV budget costs a fold surcharge of 23.8 tokens/turn (CI $[23.3, 24.4]$, 1.8% of the saving) and does not restore naive truncation's advantage. The unbounded figure to compare against Table II is 1308.6, and the two differ because that run's naive arm sits at 1557.3 rather than 1564.7 tokens/turn: the arms are the same, the session pool and turn window are not (7–39 over 12 sessions here, 7–29 over 24 in the cost run). Two surcharge figures appear below and are best fixed in one sentence: the 23.8 here is the *paired bounded-versus-unbounded* arm mean difference, whereas Table S5 (Supplementary Sec. S.M) gives 25.3 as the *fold-attributable* surcharge (209 extra tokens on the 48 fold turns amortized over 396); the 1.5-token gap between them is the bounded arm's append floor (246.7) running below the unbounded arm's mean (248.7), which the paired statistic credits but the fold-attribution does not. This turns the no-eviction assumption (T1) into a measured result. The cost ordering is also robust to load: re-firing the arms with 8 sessions contending for one shared KV pool preserves it (naive 1583 vs. stable 293 recompute tokens/turn, a paired gap within 1.5% of the single-stream figure), so the advantage is content-keyed rather than an artifact of exclusive access (T2).

S.N. Head-internal position stress and the price of the index block

Controls supporting the position and cost claims of Sec. VI-B1: the gold-line permutation inside a fixed head, and the recompute slope of the `INDEX` block measured on the published cost sessions.

a) Is the head itself immune? A position stress inside the head: That placement claim has a scope which the factorial alone does not establish, and we state the scope before the evidence. The factorial probes the value of the *newest* record, which lives in the volatile tail, so its zero says that the model reads from the region the layout designates; it does not say that where a line sits *inside* the head is irrelevant. That is the one position condition we had not measured, and it is the condition the lost-in-the-middle literature would predict to be worst [10], [11]. We therefore moved the gold line within an otherwise fixed head: 120 sessions × two head lengths (H=6, the mature operating point of the published arms, and H=24, a four-times longer head) × three slots (front, middle, back), probed at the turn where the head holds exactly H lines, one competing near-twin value left verbatim in the head in every cell, *720 calls per model on two architectures,* 0 *server errors*. The manipulation is a permutation and nothing else, verified offline before serving (`check_position_controls`): the STATE line multiset is identical across the three slots, and so is the prompt's character length: the maximum spread within a (session, H) group is 0 characters over all 240 groups, so the position contrast is read at constant total length and the slot cannot differ in how much text it carries. What varies across groups is H, and we give the conversion so a reader can judge whether the position effect could be a length effect in disguise: H=6 prompts average 1915 characters and 529 tokens (2770 and 713 at the longest), H=24 prompts average 4794 characters and 1271 tokens (6447 and 1650), so Δ2879 characters goes with Δ742 tokens, and all three slots sit inside each length. The tail block is identical. The relative order of the two TARGET lines is fixed and reported, because it decides how the collapse may be read: the competitor sits at head index 3 when gold is at the front and at index 2 in the middle and back slots (the one-position shift of removing and re-inserting gold), so gold trails the competitor by 1 line at H=6 middle, 10 at H=24 middle, 3 at H=6 back and 21 at H=24 back. The primacy slot that gold vacates is never taken by the competitor: in the 480 middle and back cells of the grid the head's first line is an auxiliary filler record, while in the 240 front cells it is the gold line itself. A loss in the middle or back cell therefore cannot be read as the near-twin winning the front of the list: it is gold's own departure from position 0, and at H=24 from both ends, since the head's last line is the gold record only in the back slot. This is a fidelity probe, not a state a prefix-stable controller can reach: an append-only head is ordered by arrival, so moving a line forward is exactly the edit that forfeits the cached prefix, so we treat it as a diagnosis of where the model reads, not as an arm of the policy space.

b) What the block costs: Fidelity alone would not settle the question: a second view is a maintenance proposal only if we can price it. We measure the recompute the `INDEX` block adds, on the *published* cost sessions rather than new ones, so the comparison is matched to the arms already in Table II. That stream covers 20 of the cost leg's 24 sessions, 460 of its 552 turns: the leg's generator records `NSESSIONS=20` as its frozen default and the width was fixed before the first call, so this is a narrower slice of the same published sessions rather than an exclusion, and the pre-emit assertion binds it byte for byte to `prefix_cost_trials.jsonl`. Consequence: this stream's reference means read 453/250.8/1,566.8 against Table II's 445.5/248.7/1,557.3, differences fully explained by the window; every premium quoted below is a within-stream paired contrast, so the window does not enter it. The generator (`gen_index_cost_trials.py`) re-derives those rows from the frozen seed and parameters and `exits` unless the `prefix_stable` and `length_only` prompts are byte-for-byte the published `prefix_cost_trials.jsonl`; it then inserts `STATE ‖INDEX` of width $I \in \{2, 8, 24\}$ live bindings ahead of the same volatile tail, and asserts that the first byte at which an index arm diverges from `prefix_stable` is exactly the end of their shared head, so a cache miss is charged only to the block, never to the state or the tail. The pre-declared prediction is a *slope*: if the block behaves as the model in Sec. V-B says, the per-turn recompute premium of `index_I` over `prefix_stable` should grow linearly in I (a fixed number of tokens per index row) and stay well under the naive window's premium, and the answer-quality rate should remain matched across all five arms so no premium can be attributed to a difficulty difference.

c) Measured price: Table S21 is the answer, and the matched-quality premise of the pre-declared prediction holds without adjustment: correctness is 1.000 in all five arms over the same 460 paired turns, so none of the premiums below can be attributed to a difficulty difference. The premium over `prefix_stable` grows with the width of the block: +27.7 tokens/turn [27.1,28.2] at I=2, +91.8 [89.5,93.9] at I=8, +195.7 [190.3,200.7] at I=24 and regressing the paired premium on I gives 7.40 tokens per `INDEX` row (SE 0.11, $p = 3.6\times10^{-24}$) with an intercept of +21.1 (SE 0.18), the block's header line, over 1,380 observations clustered on 20 sessions. The fitted line is within 11.4 tokens of every measured level, so the form that survives is the one the model in Sec. V-B predicts: a price per binding named plus a fixed frame. Two bounds on that sentence. First, three levels of I fit two parameters, so the residual degrees of freedom are zero and the form itself is not falsifiable here: the marginal price per added row falls from 10.7 (I:2→8) to 6.5 (8→24), and a saturating form would be read the same way, so what this leg establishes is monotone growth in I, compatible with $\Theta(I)$. Second, head length is fixed within this stream, so the absence of a history-length term is the separate Sec. V-B result (flat recompute under a stable head on its own sessions), not something these two-thousand-odd calls varied jointly. The reading of that slope is bounded by what this stream contains: every turn mints a fresh key ($\texttt{OBS}_t$), so no binding is ever superseded, and the premium's own dependence on I is the evidence that the block is re-prefilled from its first byte each turn (a one-row update would cost one row's tokens, not I of them). 7.40 tokens per row is therefore the price of *re-rendering* the block, an upper bound for the incremental single-row update a supersession would need, and it is measured on blocks of two to twenty-

four rows rather than on a live-binding set produced by a real key lifecycle (Sec. VII, T7). In absolute terms the $I{=}24$ block takes the prompt from 453 to 650 tokens and recompute from 250.8 to 446.6 ($\times 1.78$) while remaining $3.51\times$ below the naive window's 1,566.8; the cache hit rate falls from 44.6% to 31.3%, and prefill rises by $+28.0$ [4.2,42.0], $+66.2$ [44.0,79.7], $+117.4$ [98.5,129.3] ms, a currency carrying the run-to-run jitter quantified in Supplementary Sec. S.A(vi), which is why the claim above is stated in tokens.

The comparison that decides whether the second view belongs in the maintenance model is against the fold, not against `naive`. A bounded head that folds pays an amortized 23.8 tokens/turn (Sec. VI-B1) because its invalidation is rare, one rewrite of the whole head per $m{=}6$ appends, at $+209$ tokens on the turn it fires. A rewritable index of I bindings pays $\Theta(I)$ on *every* turn. On this workload the second view is therefore not the cheaper object: at $I{=}24$ it costs about what one fold of the bounded head costs at its worst turn, and it costs it always. What distinguishes the two terms is what they scale with: $I_{\max}$ caps the bindings a query needs named, which a workload sets, while the fold term grows with the head the history is allowed to occupy. At $I{=}2$ the premium, 27.7, is the same size as the fold's amortized surcharge, so a narrow index is affordable *alongside* a bounded head rather than instead of it, and the exact-match correctness it returns at the disadvantaged slots is the $+0.117$ to $+0.400$ of Sec. VI-D, which no setting of the fold buys.

d) Design: Three arms over the *same* 120 sessions, the same two head lengths and the same three slots as the position stress, so 720 calls per arm per model over 2 architectures, 4,320 in all from one 2,160-trial grid: `plain` is the published stress: the gold line permuted inside an append-only head; `index` renders `STATE ‖ INDEX ‖ RECENT`, where `INDEX` carries one `key -> value` line per live binding, including the queried key; `index_nok` is a placebo that carries the same block in the same position with the queried row removed, so the block's presence, framing and distance to the query are held fixed and the answer row is the only content difference. The placebo is one line shorter than the treatment rather than as long as it: with two live bindings in context the treatment block holds two rows and the placebo one, a 15-character difference. Two departures from the published stress, both declared before serving. First, a current-binding index is only well defined over distinct keys, while the published hazard deliberately lets near-twins share one key; we re-key the distractors (`RELATED`k `->` W) and, because every arm carries the re-key, the arm contrast stays internal to this run. Second, the re-key changes the prompt, so the `plain` arm is *re-measured* here rather than borrowed from Table IV: the gradient the index arm is compared against is this run's own, not a number from a different prompt. The deterministic gate (`check_index_controls`, all 2,160 trials, `GATE PASS`) asserts, before any model is called, that STATE and RECENT are byte-identical across arms and that each arm's head is a strict byte prefix of its own prompt; that the index block is *derived*, holding exactly the two live bindings present in the context in arrival order (the `index_nok` block is the same derivation with the queried key dropped, $2/1$ rows as the gate asserts, and carries no answer by construction); that the gold line keeps the same `gold_pos` in all three arms, so the index arm has the answer in *two* places, in the head where arrival put it and in the block near the end, and the only new variable is the block; that exactly one competing value sits in the head with a position shift of at most one; that the gold binding is present in every arm; and that the content is the published construction modulo the re-key. The block is small: $+70$ characters for `index` and $+55$ for `index_nok` against the 2,034-character `plain` prompt at $H{=}6$, the same $+70/+55$ at $H{=}24$ because the block does not grow with the head.

e) Pre-declared reading: Three outcomes, fixed before the first call and computed by the same shipped analyzer: **P1**, in `plain` the gradient is present (middle$-$first and last$-$first on `correct` negative and clearing the pre-declared 0.03 floor); if P1 fails this arm has nothing to fix and we say so; **P2**, in `index` the same within-arm contrasts shrink to inside the floor; **P3**, the fix is carried by the answer row rather than by a second block near the end, i.e. the difference `index_`$-$`index_nok` is positive at the disadvantaged slots while `index_nok` itself keeps the gradient. Differences are exact McNemar plus session-cluster bootstrap risk differences on the same machinery as the published legs, and the difference-in-differences $([\text{middle} - \text{first}]_{\texttt{index}} - [\text{middle} - \text{first}]_{\texttt{plain}})$, Wilcoxon-signed-rank over sessions, is the single estimand that says whether the block moves the gradient rather than shifting the level.

S.O. Content-rule leg: design, pre-declared predictions, amendments

The competitor leg's arm construction, the four predictions fixed before the first call, and the two protocol amendments disclosed in Sec. VI-B1.

a) Design: A LLMLingua-2 token classifier (`llmlingua-2-bert-base-multilingual-cased-meetingbank`, 177M parameters, CPU inference, the serving model untouched) scores the whitespace tokens of the naive window and keeps the top fraction of them [9]. Six arms over the *same* 552 (session, turn) cells, the same 24 sessions, turns 7–29, the same pinned `num_ctx`, seed, temperature and per-arm ascending-turn cache streams as the cost leg, plus `length_only` replayed inside this same invocation as a same-run control: 3,864 calls, 0 server errors. Four keep-rates $.50/.25/.20/.12$ are *open* (the answer row is not protected), and $.25/.20$ are re-run as *protected* arms (`comp_keep`*) in which the answer row is force-kept, the benefit-of-the-doubt reading a compressor's defender would ask for. Whether the queried binding survives is counted by string test in every prompt, never assumed. The deterministic gate (`check_comp_controls`, all 552 trials) asserts that each compressed prompt is a whitespace-token *subsequence* of the same trial's naive window with the probe tail byte-identical, that the budgets are monotone in the

keep-rate, that the pruner re-scores identically, and it prints the counted survival: $551/552$ at .50, $338/552$ at .25, $39/552$ at .20, $0/552$ at .12, and $552/552$ in both protected arms. The same gate measures the thing the paper is about, offline and before any call: a competitor prompt preserves ≈ 0.00 of its own previous turn's leading bytes, against 0.242 for `prefix_stable`. The two leading-byte figures in this paper are different windows and estimands and must not be interchanged: 0.242 here is the *median* longest-common-prefix share with the previous turn over the 528 consecutive pairs of the published cost stream (turns 7–29, 24 sessions), where every competitor arm reads 0.000–0.004; the 0.640 of the layout gate (Sec. IV) is the *mean* of the same shape statistic over the four 2×2 cells restricted to the mature long-head regime (from turn $V{+}9$), against 0.018 for the sliding arm. Both say the same thing about the competitor: a rewritten prompt shares almost no leading byte with its own previous turn.

b) What was fixed before the first call, and what had to be amended: Four items (`run_prefix_comp.py`, echoed verbatim by the analyzer): **C8-P1** at least one arm must reach the view's measured budget, else the leg is a size-unmatched comparison and says so; **C8-P2** the protected arm must not fall below the naive window by more than the pre-declared 0.03 floor, else pruning costs fidelity and that is the reported result; **C8-P3** the falsifiable one: no arm may recompute better than the naive window at any budget. If one did, the claim that the prefix axis is orthogonal to the content rule would be wrong; **C8-P4** at matched-or-smaller budget the view must not be weakly dominated. Two disclosures belong in the paper rather than only in the script. *Amendment 1* was written after $\sim$17% of the leg was scored: C8-P3 as pre-declared compares *absolute* recompute against the naive window, but a pruned prompt is smaller, so it recomputes fewer tokens mechanically whatever its byte stability; that contrast is confounded by B and cannot separate "sent less" from "reused more". The mechanism test is therefore the *share* R/B and the comparison at matched budget, both reported below. *Amendment 2* was written after 506–529 of the 552 turns per arm had been served, before any call of the arms it adds: the roster as first shipped only *bracketed* the view's budget (.25 above, .12 far below), while the request was for matched budget, so .20 was added with its keep-rate chosen by a fit of measured prompt tokens on kept words over the arms served so far (`matched_budget.py`). The rate choice is informed by partial observation of this leg, and the match is decided by the paired bootstrap on B, not by the fit: at the full leg's resolution the fit would pick .21 while the leg serves .20, and we report the served arm.

S.P. Discussion and threats to validity in full

The Discussion (Sec. VII) and Threats to Validity (Sec. VII) text in full. No number and no scope bound was removed from either version; the body keeps the short form of every reconciliation quoted here in full.

a) What the operating point does and does not license: Every positive result here is measured at a specific operating point: a *no-eviction*, append-only stable head replayed through a single-stream radix cache. Within that point the cost leg is large and mechanism-attributed. The layout leg comes free: the same stable placement that protects the prefix also suppresses near-twin substitution, which is why the semantic-admission lever is redundant (Sec. VI-B1). None of this transfers automatically to the regime a deployed agent actually runs in, where a finite KV budget *forces* eviction. There, the head must fold or drop within itself, the admission trade-off returns, and the redundancy between layout and admission need not hold. The paper's contribution is the constrained-view *formulation*, a mechanism-isolating *cost* result, and a *hazard characterization*; it is not an end-to-end serving benchmark or a bounded view-maintenance policy.

b) The negative result as a design-space statement: The joint-versus-decoupled null carries more information than the positive fidelity result. We built a plausible design, scoring each record by future utility minus prefix-invalidation depth and evicting jointly, to beat decoupled maintenance, and it did not: at matched budget the two levers are substitutes, and the interaction is sub-additive in both the base and tail regimes and across two families. If a stable layout already keeps the authoritative value prominent, marginal effort is better spent on the *cost* axis (making the prefix cheap to keep) than on a smarter admission score, at least until eviction binds. We report the null with its precision and with an equivalence statement rather than a bare non-significant difference: in the base regime Δ recompute is -0.22 tokens and Δ substitution -0.0025, in the tail-distractor regime -1.0 tokens (0.3%) and 0.000, and every one of those contrasts is inside the pre-declared ± 20-token and ± 0.03 margins under the session-clustered TOST of Sec. VI-C, so it is falsifiable by a future design that *does* find a regime where the levers are complements.

c) Practical reading for a system builder: For someone maintaining an agent context under a KV cache, the results support one cheap, checkable first step before any learned policy: make the reusable part of the prompt a byte-stable, append-only prefix and confine churn to a small recent tail. On our loads that single layout choice captures the recompute reduction ($6.26\times$, CI $[6.22, 6.31]$) and, in the near-twin regime, the fidelity reduction too, with no admission model, no per-record utility score, and no eviction decision. The more expensive machinery (a utility-minus-depth eviction score) is worth building only where it is demonstrably needed: once the head must fit a finite budget and content is genuinely at risk of being dropped. We measured the *timing* side of that regime (a capped head that folds, Sec. VI-B1) and it preserves the reduction, and we measured one *content* rule from outside the policy space (a token compressor, Sec. VI-B2), which is where the content side turned out to be expensive rather than cheap. What we still did not test is a rule that chooses records by their *future* value to the query, so a practitioner whose values are truly at risk should re-run a value-ranked eviction comparison before assuming the layout default carries all the way into it.

The maintenance work is also cheap in absolute terms: a saving that costs a model call to obtain would be no saving, and Table S5 (Supplementary Sec. S.E) prices `fold` at 2% of the saving it enables.

d) Why this is not just prompt ordering or rolling summarization: A fair objection to the reading above is that our controls are all *internal*: pinned, layout-only, position-permutation, and the index arm of Sec. VI-D, so none of them is a competing published policy. Two things should be said plainly. First, one obvious prefix-stable competitor *is* measured here: the budgeted arm of Sec. VI-B1 rolls the oldest head lines into a single summary line and appends, which is the structural content of a rolling summary, and the result is that rolling within a stable head keeps the reduction (1292.6 tokens/turn saved against the unbounded arm's 1316.4 in the same run) at a priced surcharge. Any policy that preserves the leading bytes inherits the mechanism; that is why we state the claim as *byte stability being load-bearing on R at matched size*, not as prefix-stability being better than every alternative. Second, the content rule is now measured as well, by an external compressor on the same sessions and turns (Sec. VI-B2), and the measurement settles the two things this paragraph previously had to leave open. At the view's own prompt budget, a token classifier that prunes without protecting the answer row is correct in 39 of 552 cells, and those 39 are exactly the cells where the binding survived the prune; force-keeping the row restores fidelity (0.998 against the naive window's 1.000, one discordant cell) and leaves the arm recomputing 170.1 [165.9,173.8] tokens more per turn than the view at statistically equal budget. Compactness buys no recompute, and what buys recompute is the leading bytes, which a content rule rewrites whatever else it decides to keep. What is still unmeasured is narrower than before and named in Sec. VI-B2: a generative summarizer, whose output replaces text rather than deleting it and so cannot be verified as a subsequence, and a rule that selects by *future* value from inside an append-only head. The practical reading is therefore still a floor to design against, not an optimum among maintenance policies.

T1: Bounded head, measured only as a fold (construct), and what the negative result does not say. The headline stable-head arm appends older records without dropping any, so on its own it never faces the admission trade-off a finite KV budget imposes. We now measure a budgeted variant that caps the head and folds the oldest digest lines into a summary (Sec. VI-B1): it keeps the head size flat and retains 1292.6 of the 1316.4 tokens/turn recompute reduction measured in that same run (98.2%; the residue is the 23.8-token surcharge that the 48 fold turns pay), so the cost conclusion does not hinge on an unbounded head. What this does not settle is the *content* of a fold: we roll up compact key/value lines, not a learned summary, and test no utility-based rule for *which* records to fold. The same bound applies to the negative result: sub-additivity of the two levers holds at the matched-budget, no-eviction point and does not imply that value-based admission is worthless under real budget pressure, which would need a bounded-eviction design. The position stress of Sec. VI-C sharpens where such a rule would first have to earn its keep: inside the head the front line dominates, so a workload whose queries target a *historical* binding, not the current one, loses the layout's protection and puts a value-ranked decision back on the table.

T2: Single-stream proxy, probed under one shared pool (cost validity). The headline recompute and latency come from an in-process llama.cpp radix cache replayed one request at a time. We push this one notch toward a serving engine by re-firing the arms with 8 sessions contending for one shared KV pool: the recompute ordering is preserved (naive 1583 vs. stable 293 tokens/turn, a paired gap within 1.5% of the single-stream figure), so the advantage is a content-keyed cache property, not an artifact of exclusive access; every arm recomputes a little more under pool pressure (stable $+18\%$), which we report. What this does *not* reach is a continuously-batched, paged-attention scheduler. Wall-clock stayed flat across arms under concurrency (≈ 8 s per request despite a $5\times$ recompute gap) because ollama serializes requests through one model slot rather than batching prefill chunks, so no throughput or TTFT separation appears. The cost result remains directional mechanism evidence; a vLLM/SGLang-class TTFT headline stays open.

T3: Construct of the quality metrics. The gold is a short synthetic code competing with meaningful passage bodies, so unforced models answer with salient body text or hedge by listing every candidate value, and *correctness* under-counts recall uniformly: it is read only under the forced single-value commit protocol, where it becomes valid (1.000 for every gold-keeping arm). *Substitution* stays construct-valid throughout and gates every fidelity claim, and the choice can neither rescue nor manufacture the joint-versus-decoupled null: joint and prefix-only tie on substitution in both protocols. On easy single-field retrieval the contamination effect is near zero, so the hazard claim is explicitly arena-scoped.

T4: Model coverage and heterogeneous magnitude. The cost effect covers `qwen3:14b` and `qwen2.5:14b` on the prefix-cost harness, plus the bounded-head replication on `qwen3:14b`; `llama3:latest` (8B) also shows the recompute reduction, but on the 2×2 and forced-commit tail-stress arms rather than the cost harness, so we count it as a mechanism replication, not a third cost measurement. The quality effect covers seven checkpoints of five server-reported architectures from 2B to 27B, one of which is not a same-generation architecture: the tag `qwen3.8:27b` is reported by the server as `qwen35`, a vision-capable architecture from a different generation than `qwen3:14b`, exercised here only through its text path. It is therefore the only 27B-level observation in the paper, and it is one checkpoint of one architecture rather than a size class sampled across families. The direction replicates in 12 of 14 strata, but the per-model magnitudes span 0.00–0.149 and neither exceed sampling noise (Q p=0.48) nor track the external size anchor (ρ=$+0.11$, exact p=0.82, leave-one-out range $[-0.29, +0.50]$, Table S12), so we report no capability law and cannot attribute the two near-null strata to scale rather

than to chance. Architecture-class independence is untested: every served checkpoint is a decoder-only transformer *on the text path we exercised*; no encoder-decoder or state-space cache was available locally to test the class. The cross-attention-design probe (`gemma2:2b`, interleaved local/global attention with qk soft-capping) preserves the direction, but it too is a decoder.

T5: Scope. Sessions are 20–30 turns at a single token budget with a five-record volatile tail.

T6: Danger-slice threshold is a proxy. The passage slice is selected by bge-m3 cosine to the naive arm's retained content, and high cosine is a weak proxy for genuine confusability: on the single-field slice 50 of 63 high-similarity trials did *not* contaminate at temp 0. We mitigate by reporting the effect on the harder passage regime where the hazard is actually live, by scanning several thresholds (Table S10), and by scoping the claim to direction-consistency and the pooled estimate rather than to any single near-null stratum.

T7: The `INDEX` view is measured as a rendering and a price, not as a maintenance algorithm. Its two legs probe single-turn prompts, and neither leg generates the workload the view is ultimately for. In the fidelity grid the block holds exactly two rows in every one of its 720 cells per architecture (the queried binding and one re-keyed near-twin), so nothing about a large live-binding set is observed; in the cost stream every turn mints a fresh key ($\texttt{OBS}_t$), so no key is ever re-bound and the block, re-rendered in full each turn, is priced as a *whole-block rewrite*, an upper bound on the per-turn premium, not the cost of updating one row. Two decisions the algebra needs are consequently specified in Sec. IV-A but untested here: which bindings count as live (we render the I most recent; reference-reachability and session TTL policies are not run), and what is evicted when live keys exceed $I_{\max}$ (no overflow occurs in either leg, so the eviction rule's fidelity cost is unmeasured). The third is now measured: served back on the shared-key `STATE`, where `STATE` and `INDEX` disagree about `TARGET`, the block still recovers the collapse and a stale index that names the competitor drives substitution above the raw state (Table S20), so the model reads the view over the line beneath it. The claim C7 supports is therefore: a rewritable near-end block relocates the recompute price from history length to binding count, shrinks the in-head position gradient without removing it, and holds under the shared-key hazard that originally motivated it. Whether the block pays for itself under a real binding lifecycle is a separate experiment.

T8: One external content rule, at one budget, with a fit-chosen rate. The competitor of Sec. VI-B2 is a single published token-level compressor [9] (a 177M-parameter relevance classifier) applied at whitespace-word granularity, and its 512-position encoder forces the 480-word chunking described there, so each arm is a word-keep fraction under one deterministic rank order, not a general span-selection policy. The budget it is matched to is the append-only view's *mean* prompt size on these sessions: nothing is claimed about a compressor tuned to another budget, and the three arms at or below the view's budget (`comp_open20` 434.2, `comp_open12` 291.7, `comp_keep20` 442.6 against 445.5) are where the dominance test returns no relation rather than a loss. The matched rate itself is the design's most post-hoc element, disclosed as Amendment 2: it was picked by regressing measured prompt tokens on kept words over arms partially served in this very leg, and at the full leg's resolution that fit names .21, an arm never served, so the paper reports the arm actually served and lets the paired bootstrap on B adjudicate the match. Two content rules remain untested and are the honest answer to "have you compared to the state of the art?": a *generative* summarizer, whose fidelity cost depends on which records a language model chooses to keep, and a rule that ranks records by *future value* inside an append-only head; the second is the joint policy whose decoupled version is the null result C5, and the first would rewrite the front by construction. Finally, the gold-protected arms force-keep the answer row, which is benefit of the doubt rather than a deployable policy for a system that cannot know in advance which record the next query reads.

Net. The positive claims are bounded to a cacheable-prefix, no-eviction, single-stream operating point and to near-twin-rich passage retrieval; the negative claim is bounded to the matched budget tested; the second view's claim is bounded to a two-row block with no superseded key; the external comparison is bounded to one compressor at one budget (T8). Each is stated at that scope, and none is presented as an end-to-end serving result.

S.Q. Evaluation setup detail in full

The text of Sec. VI in full. Every number, threshold and scope bound below also appears in the body.

a) Data and metrics: Cost measurements use real LongBench-derived multi-turn histories replayed through the serving proxy [8]. Quality measurements use (i) a passage-retrieval layout adapted from LongBench [8], where competing near-twin values are retained in the volatile tail, and (ii) a controlled multi-turn harness with synthetic gold codes on a single-field retrieval regime used as the "easy" contrast. Session lengths are 20–30 turns at a single token budget with a five-record volatile tail.

The passage danger slice is drawn by a prevalence scan rather than by hand: every candidate distractor–gold pair is embedded with the BGE M3 model (`bge-m3`) [12], thresholded on the naive arm's own retained similarity ($\texttt{lo_max} \geq 0.6$), and kept only where naive positional truncation actually *retains* a near-adversary passage, the regime in which the hazard can bite. This yields 300 usable questions ($K{=}4$ distractors, truncation 500 tokens), of which ≈ 47 form each per-config paired stratum (44 for the two 27B strata, which read the shorter of the two danger trial files; the three items that file lacks are `2wikimqa`, and they are recorded rather than explained in Sec. S.E. Nothing about the pool rule differs between them: the two 27B strata enter the pre-specified 14-stratum pooling of C4 on their own 44 items. The single-field MTU-style slice (345 groups, threshold 0.52) is retained

precisely to report its *null*: on easy retrieval the co-present-and-confusable regime is rare, which is why the hazard claim is arena-scoped rather than a blanket "long context is unsafe" statement.

Fidelity is *substitution* (fraction of answers that report a competing near-twin rather than the gold) and, only under the forced single-value protocol, *correctness*. Cost is per-turn *recompute* (tokens) and *prefill latency* (ms), both read from the serving proxy counters. Bloat is tokens sent per turn.

b) Serving environment and measurement validity: All runs use a single local `ollama` server backed by the llama.cpp radix prefix cache; the cost legs pin `num_ctx` to 8192, while the single-turn fidelity slices send prompts of at most 1,072 tokens with no override, so no stratum in this paper approached a context limit. Sessions are replayed sequentially as one request per turn, each arm its own ascending-turn cache stream so the prefix warms exactly as in a live conversation. Recompute and prefill are read from the response's `prompt_eval_count`, `prompt_eval_cached_count` and `prompt_eval_duration`. Every cost number rests on those fields, so `verify_cache_measure.py` checks them before any cost claim is read, persisting six checks with their per-prompt events (Supplementary Sec. S.A): the field behaves as a *token*-level ruler rather than a whole-prompt flag; on a constructed prefix the cached count matches the designed prefix length and the recompute is the changed suffix and nothing more; a warmed prefix survives interleaved load at our scale, so a low hit in these runs is layout, not cache pressure; engine version, GPU count and `num_ctx` are recorded and the embedded llama.cpp build, whose commit the CLI does not expose, is pinned by the SHA-256 of the server binary and of the bundled llama libraries rather than guessed; the prefill-duration field is reported only as a proxy, with its run-to-run spread measured on this shared-GPU box (check vi); and the anti-reuse nonce is the *first* prompt line, constant within an arm's stream, so it can neither manufacture a cross-run whole-prompt hit nor move the front bytes an arm is meant to share. This is a single-stream proxy, not a batched/paged engine; the boundary is drawn as T2.

c) Statistics and decision thresholds: Cost gaps are reported with session-cluster bootstrap intervals, the cluster being the unit the design randomizes over; paired fidelity contrasts use the exact McNemar test on discordant turn-pairs [13], and pooled quality effects across model families and temperatures use a Mantel–Haenszel test [14] with a question-cluster bootstrap risk difference, so a repeated item contributes one resampled unit rather than one per arm. The pooled stratum set is our pre-specified primary analysis (declared at revision, not registered before the runs): every served (model, temperature) cell with no capability or significance filter, a size-filtered pool appearing only as a labelled subgroup analysis (Section VI-B1) so the pooling rule cannot be chosen after seeing which layers look favourable. Cross-model effect *size* claims are tested against an external anchor rather than asserted from a table, and the sized control isolates mechanism from compactness. Supplementary Sec. S.B gives the independent-temperature-0 statistic, the permutation test, the hazard logistic model, the resample counts and the pooling audit.

A maintenance policy is scored as an *actionable* improvement only if it lowers substitution by at least 0.03 *and* recompute by at least 20 tokens at matched quality, a pre-declared smallest effect size of interest, not a statistical hair [15], declared before the confirmatory runs and not revised afterwards. The floor alone does not license an equality claim: a difference that misses it is *undetermined*, and only a session-clustered two one-sided tests (TOST) procedure against ± 0.03 and ± 20 turns "below the floor" into "equivalent within the declared margins" [1]. Supplementary Sec. S.C records how each floor is anchored to a measured quantity and the audit trail of that re-analysis.

d) Sample flow, model roster and call budget: A reported n is only interpretable if the reader can see what was dropped and why, so Supplementary Table S4 (Sec. S.E) traces each multi-turn study from generated records to analysed cells, per arm, computed from the run files rather than typed (`sample_flow.py`). The analysis window is a *turn range fixed by the generator before any model call* and the same rule for every arm of a study, so designed cells equal calls issued equal cells analysed and there is no post-hoc exclusion. The confirmatory analyses of this paper rest on 34,134 calls in 38 named run files, all of them counted; one further file (276 calls) is the latency-replication diagnostic of Supplementary Sec. S.A, kept in the audit so that no call goes uncounted (34,410 calls in 39 files in all) but carrying no inferential claim. Four legs are itemised separately in the audit: rewritable-index fidelity, index cost, the external content rule of Sec. VI-B2, and the shared-key re-servation of the index behind Table S20, and their call counts, arm rosters and the trial-versus-call arithmetic of each grid are accounted in Sec. S.E.

All models are served locally through the same `ollama`/llama.cpp radix prefix cache; no cloud endpoint is used. They are released checkpoints of publicly documented families [3], [4], [5], [6], [7], named by the tags the server reports: two 14B architectures for the cost harness, seven checkpoints from 2B to 27B spanning five distinct server-reported architectures (`ollama show` reports `mistral:latest` as `llama`, the same architecture as `llama3:latest`, and reports both `qwen3:14b` and `qwen3:8b` as `qwen3`; the environment file records each digest) at two temperatures for the fidelity passage sweep, one architecture for the 2×2 and layout factorial, and a second with a replicate epoch for the forced-commit tail stress. The single 27B checkpoint is not a same-generation architecture: the server reports `qwen3.8:27b` as `qwen35`, a vision-capable design exercised here through its text path only, so the 27B end of the range is one checkpoint of one architecture rather than a sampled size class (Sec. VII, T4). Supplementary Table S1 lists each model with its exact per-experiment call count over the 38 confirmatory files, and names the 39th file separately as the check vi replication.

```
Naive (sliding window, recent-first)
QUERY: current value of ORDER_ID?
RECENT TOOL RESULTS
turn3: search(···) -> ORDER_ID=A-77X
turn2: create(···) -> ORDER_ID=A-77
turn1: get_cust(···) -> cust=Northwind
Stable head (append-only)
QUERY: current value of ORDER_ID?
STATE (accumulated)
ORDER_ID=A-77 //sealed, not rewritten
RECENT TOOL RESULTS
turn3: search(···) -> ORDER_ID=A-77X
```

Fig. S1. Worked rendering of one session. Folded digest lines (left column of the `STATE` block) are written once and never edited, so the prefix ℓ_t grows monotonically; the raw `RECENT` block is the only part that changes. Content is illustrative of the template; measured values in Sec. VI-B1 use real LongBench record bodies.

e) Reproducibility and experimental matrix: Each experiment is a deterministic replay of a generated session set. Randomness is fixed by seed (`1234`/`777`) and temperature (0 and 0.5–0.7); *per-config and per-run nonces* are injected into each arm's prompt so a stale whole-prompt radix node from a prior invocation can never be credited as a cache hit. Record content is always real LongBench text [8], of two different materials detailed in Supplementary Sec. S.D: which harness takes multi-document QA items directly and which reassembles sentence bodies into constructed append-only sessions. The fidelity floor (gold present in every arm and turn) is asserted before any substitution claim is read, and `correct` is used only under the forced single-value commit protocol, where it is construct-valid. Supplementary Table S2 names the script triple behind every number and Table S3 collects every run on one axis set: what it manipulates, model coverage, sample size, endpoint, and which claim it feeds, so a reader can trace each result card to a concrete experiment; per-prompt event logs (`*.jsonl`/`*.json`) sit beside each write-up in the code package. One command, `python headline_numbers.py`, re-reads those logs and recomputes every mean, ratio, risk difference, confidence interval and p-value quoted in this paper, including the cells of the position and index tables, so a transcription slip in the prose shows up as a disagreement with that output rather than surviving in print. The package (code, gates, per-call logs and regenerators) ships with this submission as an anonymised artefact.

S.R. Supplementary figures

The figures that carry measured results sit in the body, where the bars and the numbers have to be read together; the one figure that would not survive that move is kept here. Figure S1 is the object all those counters are taken on—the template of Sec. IV-A rendered for a single order-id lookup under both arms—and it is checkable without a model call: the stable arm's leading bytes name the same turn at every request, the naive arm's name a different one, which is the whole difference the recompute counters then price.

References

[1] D. Lakens, A. M. Scheel, and P. M. Isager, "Equivalence testing for psychological research: A tutorial," *Advances in Methods and Practices in Psychological Science*, vol. 1, pp. 259–269, 2018.

[2] Z. Yang, P. Qi, S. Zhang *et al.*, "Hotpotqa: A dataset for diverse, explainable multi-hop question answering," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018, pp. 2369–2380.

[3] A. Yang, A. Li, B. Yang *et al.*, "Qwen3 technical report," 2025. [Online]. Available: http://arxiv.org/abs/2505.09388

[4] A. Yang, B. Yang, B. Zhang *et al.*, "Qwen2.5 technical report," 2024. [Online]. Available: http://arxiv.org/abs/2412.15115

[5] A. Grattafiori, A. Dubey, A. Jauhri *et al.*, "The llama 3 herd of models," 2024. [Online]. Available: http://arxiv.org/abs/2407.21783

[6] Gemma Team, M. Riviere, S. Pathak *et al.*, "Gemma 2: Improving open language models at a practical size," 2024. [Online]. Available: http://arxiv.org/abs/2408.00118

[7] A. Q. Jiang, A. Sablayrolles, A. Mensch *et al.*, "Mistral 7b," 2023. [Online]. Available: http://arxiv.org/abs/2310.06825

[8] Y. Bai, X. Lv, J. Zhang *et al.*, "Longbench: A bilingual, multitask benchmark for long context understanding," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024, pp. 3119–3137.

[9] Z. Pan, Q. Wu, H. Jiang *et al.*, "Llmlingua-2: Data distillation for efficient and faithful task-agnostic prompt compression," 2024. [Online]. Available: http://arxiv.org/abs/2403.12968

[10] N. Salvatore, H. Wang, and Q. Zhang, "Lost in the middle: An emergent property from information retrieval demands in llms," 2025. [Online]. Available: http://arxiv.org/abs/2510.10276

[11] J. Gabín, A. Perez, and J. Parapar, "Lost in the evidence? reproducing document position and context size effects in rag," 2026. [Online]. Available: http://arxiv.org/abs/2605.27105

[12] J. Chen, S. Xiao, P. Zhang *et al.*, "M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," 2024. [Online]. Available: http://arxiv.org/abs/2402.03216

[13] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, pp. 153–157, 1947.

[14] N. Mantel and W. Haenszel, "Statistical aspects of the analysis of data from retrospective studies of disease," *JNCI: Journal of the National Cancer Institute*, vol. 22, no. 4, 1959.

[15] F. Anvari and D. Lakens, "Using anchor-based methods to determine the smallest effect size of interest," *Journal of Experimental Social Psychology*, vol. 96, p. 104159, 2021.

TABLE S2

RUN MANIFEST. FOR EACH STUDY, THE SCRIPT THAT GENERATES ITS SESSIONS, THE ONE THAT ISSUES THE CALLS, AND THE ONE THAT COMPUTES THE REPORTED STATISTICS. THE PARENTHESISED FILE IS THE PROSE WRITE-UP OF THAT COMPUTATION; THE MACHINE-WRITTEN ARTEFACTS ARE THE PER-PROMPT EVENT LOGS (`*SCORED*.JSONL`, ONE ROW PER CALL WITH ITS SERVER COUNTERS) AND `HEADLINE_NUMBERS.{PY,JSON}`, WHICH REGENERATES EVERY RATIO, CI AND p-VALUE QUOTED IN THIS PAPER FROM THOSE LOGS IN ONE PASS. SCRIPT NAMES APPEAR WITHOUT THE `.PY` SUFFIX.

Study (claim)	**generate → run → analyse (report)**
Cost, 3 arms (C2)	gen_prefix_cost → run_prefix_cost → analyze_prefix_cost (`prefix_cost_report.md`)
Bounded head (C2)	gen_prefix_bounded → run_prefix_bounded → analyze_prefix_bounded (`prefix_bounded_report.md`)
Joint 2 × 2 (C5)	gen_joint_sessions → run_joint_2x2 → analyze_joint, analyze_2x2_correct (`joint_2x2_report.md`)
Layout 2 × 2 (C5)	gen_joint_sessions → run_joint_layout → check_joint_layout, analyze_layout, check_layout_controls (`joint_layout_report.md`)
Passage danger (C4)	gen_lb_danger → run_lb_danger → analyze_lb_danger, analyze_pooled (`lb_danger_report.md`); `analyze_pooled` takes the pool as an argument (no argument is the pre-specified 14-stratum primary, `size` the size-only sensitivity pool, `subgroup` the superseded outcome-filtered pool) and prints both T and the bootstrap (writes `pooled_all.txt`, `pooled_size.txt`, `pooled_subgroup.txt`); analyze_capability (`capability_analysis.txt`), power_mde (the 80%-power minimum detectable effect and the n needed to clear the 0.03 floor)
Field danger (C3)	gen_danger_trials → run_danger_infer → analyze_danger (`danger_report.md`)
Hazard model (C3)	gen_policy_trials → run_policy_infer → analyze_policy (`policy_results.md`)
Position stress (C6)	gen_position_trials → run_position_stress → analyze_position (`position_analysis.txt`, `position_analysis.json`); check_position_controls the pre-run gate
Index view (C7)	fidelity: gen_index_trials → run_index_stress → analyze_index (`index_analysis.txt`), with check_index_controls as the gate; cost: gen_index_cost_trials → run_index_cost → analyze_index_cost (`index_cost_analysis.txt`); shared-key re-servation: gen_index_sk_trials → check_index_sk_controls → run_index_sk_leg → analyze_index_sk, with headline_numbers.shared_key_leg recomputing the contrasts. make_index_tables writes the float files the manuscript reads verbatim, so no cell of the index tables (fidelity, cost or shared-key) is typed; its post-hoc answer-form split calls the classifier in `diag_answer_form`
External content rule (C8)	compress_llmlingua2 (prunes the published cost-leg windows; no new sessions) → run_prefix_comp → analyze_prefix_comp, make_comp_table (`comp_analysis_summary.json`, `comp_table.tex`); check_comp_controls is the pre-run gate (subsequence, monotone budgets, deterministic rescore, counted gold survival), and matched_budget computes the keep-rate from the run files, reporting the fit's pick beside the arm the leg actually served
Cache validity (Sec. VI-A)	probe_cache_field, verify_cache_measure, summarize_cache_verify, sample_flow, repeat_ms_variance (`cache_verify_report.json`, `sample_flow.md`, `ms_repeat_variance.json`)
Fold ledger (Sec. V)	fold_cost (reads the bounded run files; Table S5)
Concurrency probe (Sec. VII)	run_prefix_concurrency (`--parallel 8`), whose printed summary is `prefix_concurrency_report.md`
Headline numbers	headline_numbers (regenerates every CI and p-value quoted here)

TABLE S3

EXPERIMENTAL MATRIX. EACH RUN, ITS MANIPULATED AXIS, MODEL COVERAGE, SAMPLE SIZE, PRIMARY ENDPOINT, AND THE CLAIM IT SUPPORTS. n IS PAIRED TURNS/QUESTIONS/CELLS AS NOTED; EVERY COUNT IS AUDITED IN TABLE S4 AND EVERY FILE IS NAMED IN TABLE S2.

Run	Axis	n	Endpoint	Claim
Cost, 3 arms	prefix stability	552 turns	recompute, prefill	C2
Cost, 2nd architecture	replication	552 turns	recompute, prefill	C2
Bounded head (3 arms)	head budget B_{KV}	396 turns	recompute, fold cost	C2
Joint 2 × 2	distract. × depth	408/cfg	subst., recompute	C5
Layout 2 × 2	label × stab.	408/cell	subst., recompute	C5
Tail stress (forced)	depth × tail tw.	120×6	subst., correct	C5
2nd architecture	replication	120×4	subst., correct	C5
Gold-position stress	slot in head $\times H$	120×6	correct, subst.	C6 scope
2nd architecture	replication	120×6	correct, subst.	C6 scope
Index fidelity (3 arms)	`INDEX` block × slot $\times H$	120×6	correct, subst.	C7
2nd architecture	replication	120×6	correct, subst.	C7
Index cost (5 arms)	index breadth $I \in \{2, 8, 24\}$	460 turns	recompute, prefill	C7
Shared-key index reserv.	shared-key conflict × slot $\times H$	120×6	correct, subst.	C7
Content rule, 7 arms	keep-rate × gold protection	552 turns	recompute at matched B	C8
2nd architecture	replication	552 turns	recompute at matched B	C8
Passage danger	naive vs M-A	47/q×14	substitution	C4
Field (single-value) danger	naive vs M-A	90–91/arm	substitution (null)	C3
Hazard model	max_sim, K	270	logistic β	C3
Cache-validity probes	counter semantics	272 cache, 276 repeat	token-level reuse	C2
Concurrency probe	shared KV pool	120 calls	recompute invariance	C2

TABLE S4

SAMPLE FLOW OF EVERY STUDY THAT ISSUES CALLS IN A GRID OR A STREAM, COMPUTED FROM THE RUN FILES. "DESIGNED" IS THE CELL COUNT IMPLIED BY THE PRE-DECLARED TURN WINDOW (STREAM ROWS) OR BY THE FROZEN GRID (PROBE ROWS, MARKED ‡), IDENTICAL FOR EVERY ARM OF A STUDY; "ISSUED"/"ANALYSED" ARE WHAT THE SERVER ACTUALLY RETURNED AND WHAT ENTERS THE ANALYSIS. NO STUDY HAS ANY POST-HOC EXCLUSION.

Study (model)	arms	gen.	designed	issued	err.	analysed
Cost (qwen3:14b)	3	720	552	552	0	552
Cost (qwen2.5:14b)	3	720	552	552	0	552
2×2 base	4	480	408	408	0	408
Layout 2×2	4	480	408	408	0	408
Tail stress (forced)	4	480	120	120	0	120
Tail stress, llama3	4	480	120	120	0	120
Bounded head	3	480	396	396	0	396
Index cost (5 arms)	5	460	460	460	0	460
Content rule (qwen3:14b)	7	552	552	552	0	552
2nd architecture	7	552	552	552	0	552
Gold-position stress‡, qwen3	3	240	240	240	0	240
llama3‡	3	240	240	240	0	240
Index fidelity‡, qwen3	3	720	720	720	0	720
llama3‡	3	720	720	720	0	720
Shared-key index‡, qwen3	2	720	720	720	0	720
llama3‡	2	720	720	720	0	720
Latency replication†	2	138	138	138	0	138

Turn windows used: 7–29 (cost, index cost, content rule: the competitor leg replays the published cost sessions on their own pre-declared window), 3–19 (2×2, layout), 15–19 (tail stress), 7–39 (bounded head, 12 sessions, the arm that answers the finite-budget objection in Sec. VII), 7–29 $\times 3$ repeats (replication, 2 sessions). The ‡ rows are single-turn probe grids: one call per session$\times$cell (120 sessions $\times$ 3 slots $\times$ 2 head lengths for position, the same 720-cell grid $\times$ 3 arms for index fidelity and $\times$ 2 arms for the shared-key re-servation, where the distractor is re-keyed `TARGET` so a stale value competes under one key), issued from the frozen gold-position session pool. Per-study maxima run 998–1870 tokens; the longest prompt in any prefix-stable arm is 744 (9.1% of `num_ctx`), in the widest rewritable index arm `index_I24` 922 (11.3%), in the probe grids 1677 (20.5%) and in any naive or naive-window arm 1870 (22.8%), so nothing in this table could have been dropped for overflowing the pinned window. This table and Table S1 together are the full inventory of the studies that issue calls, and every row of both enters the 34,134-call confirmatory audit of Sec. VI-A. †The last row is the measurement-validity replication of check vi: the same 276 calls, counted in the 34,410-call 39-file audit but outside the 34,134 confirmatory calls, because it re-fires published prompts and tests no hypothesis.

TABLE S5

COST LEDGER OF THE MAINTENANCE OPERATOR ITSELF (SEC. V), SO THE REPORTED PREFILL SAVING IS A NET NUMBER. CPU IS TIMED ON THE EXPERIMENT HOST AND REPEATS TO WITHIN A FEW PERCENT; THE TOKEN ROWS ARE READ OFF THE BOUNDED-ARM RUN FILE AND DO NOT VARY. THE LAST ROW IS DELIBERATELY *not* A MEASUREMENT. EACH FOLDED HEAD LINE IS ONE `KEY=VAL` ROW AND NOTHING ELSE (`OBS0=ALPHA0` THROUGH `OBS28=ECHO28` IN THE SERVED STREAM, 10–15 CHARACTERS, THE RAW SENTENCE BODY DROPPED AT THE FOLD): THAT IS WHAT THE $\Delta \approx 9$ TOKENS/LINE OF SEC. V-B PRICES.

Quantity	**Value**
Reference fold (append one `key=val` row)	≈ 0.15 ms/op over 4000 ops (0.6 s); 0 model calls, 0 output tokens
Bounded-head roll-up (measured arm)	48 folds in 396 turns; 0.3 μs mean, 1.2 μs p95, 3.4 μs max; < 1 ms for the whole stream
Model calls / output tokens from folding	0 / 0
Recompute surcharge of a fold turn	recompute 456 tok vs 247 on an append turn (+209), on 48 of 396 turns
amortized	25.3 recompute tok/turn, 2.0% of the saving; 23.8 net against the unbounded arm (Sec. S.M)
Measured saving the fold must beat	1293 tok/turn (naive 1565 − bounded 272)
Break-even ceiling for an *LLM*-based fold	$\leq$ 1293 recompute tokens/turn *and* no fidelity loss, **not measured**: no arm calls a model to fold

TABLE S6

COST ATTRIBUTION (C2): SESSION-CLUSTER BOOTSTRAP OF THE MEAN PAIRED RECOMPUTE DIFFERENCE (TOKENS, `QWEN3:14B`). THE SIZE-MATCHED `PINNED_RAW` CONTROL REPRODUCES 97% OF THE NAIVE→M-A GAP, ISOLATING PREFIX INVALIDATION FROM COMPACTNESS. RUN: COST STUDY (TABLE S2).

Contrast	D	95% CI	verdict
LO − M-A	1308.6	$[1280.1, 1331.0]$	SIG
LO − pinned	1272.1	$[1243.9, 1294.2]$	SIG
M-A − pinned	-36.5	$[-37.8, -35.2]$	pinned more
LO − M-A (ms)	850.8	$[826.9, 868.2]$	SIG
LO − pinned (ms)	825.8	$[801.1, 843.4]$	SIG

TABLE S7

FACTORIAL EFFECTS (C5): MAIN EFFECTS AND INTERACTION ON THE TWO AXES, WITH SESSION-CLUSTER BOOTSTRAP 95% CIS (`QWEN3:14B`, 4000 RESAMPLES). BOTH MAIN EFFECTS ARE LARGE AND SIGNIFICANT; THE SUBSTITUTION INTERACTION IS *sub-additive* (CI EXCLUDES ZERO ON THE POSITIVE SIDE), I.E. THE SEMANTIC LEVER'S BENEFIT IS LARGE UNDER A SLIDING HEAD AND ≈ 0 UNDER A STABLE HEAD: REDUNDANCY, NOT SYNERGY. THE RECOMPUTE INTERACTION IS NOISE IN BOTH REGIMES. VALUES VERBATIM FROM THE REPORT.

Regime	Effect	Δ	95% CI
Base	$d \to$ subst ($c_{10} - c_{00}$)	-0.392	$[-0.495, -0.294]$
Base	$p \to$ recom ($c_{01} - c_{00}$)	-262.8	$[-276, -249]$
Base	subst interaction	$+0.390$	$[+0.299, +0.488]$
Base	recom interaction	$+0.47$	—
Tail	$d \to$ subst ($c_{10} - c_{00}$)	-0.808	$[-0.883, -0.725]$
Tail	$p \to$ recom ($c_{01} - c_{00}$)	-412.9	$[-429, -396]$
Tail	subst interaction	$+0.808$	$[+0.733, +0.883]$
Tail	recom interaction	$+0.14$	—

TABLE S8

DANGER-SLICE PREVALENCE VS THRESHOLD (C4), QWEN PASSAGE POOL. OF 300 USABLE QUESTIONS (K=4, TRUNC=500), THE COUNT WHOSE NAIVE WINDOW RETAINS A PASSAGE AT COSINE $\geq$ THE THRESHOLD. WE REPORT THE COUNT AT *every* THRESHOLD RATHER THAN ONLY THE OPERATING POINT (BOLD, THE ANALYSIS SLICE), SO THE SLICE IS NOT CHOSEN POST HOC; IT SHRINKS $74 \to 3$ AS THE BAR RISES, AND THE POOLED DIRECTION IN TABLE S10 IS STABLE ACROSS IT. SOURCE: THE SAME PASSAGE-DANGER RUN FILES (TABLE S2).

thr $\geq$	0.55	**0.60**	0.65	0.70	0.75	0.80
questions	74	**47**	23	14	9	3

TABLE S9

CONTAMINATION HAZARD MODEL (C3): LOGISTIC REFIT ON NATURAL FIELD VALUES, CONTAMINATION $\sim$ MAX_SIM $+K$, FIT ON THE *three-arm* POLICY RUN (`QWEN3:8B`, TEMP 0; n=270 TRIALS = 90 PER ARM, 13 CONTAMINATION EVENTS = 4.8%; TABLE S2, ROW "HAZARD MODEL"). THIS IS A *different* RUN FROM THE SINGLE-FIELD PAIRED DANGER SLICE (91 ELIGIBLE PAIRS/ARM, ALL-ZERO DISCORDANCE, TABLE S13): THE NULL LIVES ON THE SINGLE-FIELD SLICE, WHEREAS THE HAZARD IS ESTIMATED ON THE THREE-ARM NATURAL FIELD WHERE MAX_SIM SPANS 0.22–0.74. SEMANTIC PROXIMITY TO THE GOLD PREDICTS CONTAMINATION; K (DISTRACTOR COUNT) DOES NOT. THE 13 EVENTS FALL ENTIRELY ABOVE THE τ_{NEAR}=0.45 BAND (0.102 VS 0.000 CONTAMINATION RATE ABOVE/BELOW IT), SO THE INTERCEPT/β PAIR ($-8.37, 8.16$) SITS IN A NEAR-SEPARATION REGIME AND THE UNPENALISED β OVERSTATES THE EFFECT SIZE; WE THEREFORE ALSO REPORT AN L2-PENALISED REFIT AND THE DISCRIMINATION, WHICH ARE STABLE: SHRINKAGE $L2$ ($\lambda\approx1$) GIVES β_{SIM}=1.27, AND THE ROC AUC IS 0.74 (PENALISED) TO 0.81 (UNPENALISED). THE CLAIM WE MAKE IS THAT *proximity orders the risk* (POSITIVE, MONOTONE, AUC WELL ABOVE 0.5), NOT THAT THE COEFFICIENT EQUALS 8.16. MCFADDEN ρ^2=0.145 IS A PSEUDO-R^2 AT A 4.8% EVENT RATE.

Term	β (unpen.)	p (FDR-adj.)	β ($L2$)
intercept	-8.372	—	-4.73
max_sim	**8.161**	**0.001**	**1.273**
K	0.552	0.165	0.534
n / events	270 / 13 (4.8%), `qwen3:8b`, 3-arm run		
McFadden ρ^2	0.145	AUC 0.74–0.81	
VIF(max_sim| K)	1.04	events all at max_sim>0.45	

TABLE S10

PASSAGE DANGER SLICE (C4): SUBSTITUTION, NAIVE VS M-A, ACROSS 14 CONFIGURATIONS (LONGBENCH-DERIVED PASSAGES, THR≥ 0.6, n PAIRED QUESTIONS). WILSON 95% CIS; MCNEMAR b (LO SUBST, M-A CLEAN) / c (M-A SUBST, LO CLEAN), EXACT p; AND, BETWEEN THEM, Δ SUBST $= (b-c)/n$ WITH ITS 95% NORMAL-APPROXIMATION INTERVAL $\Delta \pm 1.96\sqrt{b+c}/n$, THE QUANTITY BODY FIG. 5 DRAWS (n=47, OR 44 FOR THE TWO `QWEN3.8:27B` STRATA). DIRECTION (M-A LOWER) HOLDS STRICTLY IN 12/14 AND TIES IN THE TWO `QWEN3:8B` STRATA; ONLY `QWEN2.5:14B` t=0 IS INDIVIDUALLY SIGNIFICANT; THE 2B AND 8B STRATA ARE NEAR-NULL. PRIMARY POOL OVER ALL 14 STRATA: MANTEL–HAENSZEL T=18.99, p=1.3×10^{-5} (7 INDEPENDENT TEMPERATURE-0 STRATA: T=10.88, p=9.7×10^{-4}); QUESTION-CLUSTER BOOTSTRAP $D = 0.0782$, CI $[+0.0031, +0.1631]$, $P(D \leq 0) = 0.021$. SIZE-ONLY SENSITIVITY (THE 12 STRATA OF CHECKPOINTS $\geq$ 7B, NO OUTCOME FILTER): T=22.15, p=2.5×10^{-6}, $D = 0.0860$, CI $[+0.0144, +0.1679]$. RUN FILES: TABLE S2, ROW "PASSAGE DANGER".

Model	t	LO subst [CI]	M-A subst [CI]	b/c	Δ subst [95%]	p
gemma2:2b	0.0	0.383 [.258,.526]	0.340 [.222,.483]	9/7	0.043 $[-0.124, +0.210]$	0.804
gemma2:2b	0.6	0.362 [.240,.505]	0.340 [.222,.483]	9/8	0.022 $[-0.150, +0.194]$	1.00
llama3	0.0	0.213 [.120,.349]	0.128 [.060,.252]	7/3	0.085 $[-0.047, +0.217]$	0.344
llama3	0.6	0.213 [.120,.349]	0.128 [.060,.252]	7/3	0.085 $[-0.047, +0.217]$	0.344
mistral	0.0	0.234 [.136,.372]	0.085 [.034,.199]	10/3	0.149 $[-0.001, +0.299]$	0.092
mistral	0.6	0.234 [.136,.372]	0.085 [.034,.199]	9/2	0.149 $[+0.011, +0.287]$	0.065
qwen2.5:14b	0.0	0.149 [.074,.277]	0.021 [.004,.111]	6/0	0.128 $[+0.026, +0.230]$	**0.031**
qwen2.5:14b	0.6	0.149 [.074,.277]	0.064 [.022,.172]	6/2	0.085 $[-0.033, +0.203]$	0.289
qwen3.8:27b	0.0	0.136 [.064,.267]	0.068 [.023,.182]	6/3	0.068 $[-0.066, +0.202]$	0.508
qwen3.8:27b	0.5	0.159 [.079,.294]	0.068 [.023,.182]	7/3	0.091 $[-0.050, +0.232]$	0.344
qwen3:14b	0.0	0.234 [.136,.372]	0.128 [.060,.252]	7/2	0.106 $[-0.019, +0.231]$	0.180
qwen3:14b	0.6	0.234 [.136,.372]	0.149 [.074,.277]	7/3	0.085 $[-0.047, +0.217]$	0.344
qwen3:8b	0.0	0.170 [.089,.301]	0.170 [.089,.301]	2/2	0.000 $[-0.083, +0.083]$	1.00
qwen3:8b	0.7	0.170 [.089,.301]	0.170 [.089,.301]	2/2	0.000 $[-0.083, +0.083]$	1.00

TABLE S11

ROBUSTNESS OF THE POOLED DANGER-SLICE EFFECT (C4). THE PRIMARY POOL IS EVERY SERVED (MODEL, TEMPERATURE) STRATUM, OUR PRE-SPECIFIED PRIMARY ANALYSIS (DECLARED AT REVISION); THE SENSITIVITY POOL IS EVERY SERVED CHECKPOINT OF $\geq$ 7B PARAMETERS (12 STRATA, 6 INDEPENDENT MODELS), A CRITERION FIXED WITHOUT LOOKING AT b OR c. ADDING THE TWO `GEMMA2:2B` STRATA BACK DILUTES THE MANTEL–HAENSZEL STATISTIC FROM 22.15 TO 18.99, BUT THE DIRECTION AND BOTH CLUSTER-BOOTSTRAP INTERVALS SURVIVE; THE EFFECT IS NOT AN ARTIFACT OF POOLING ONLY GQA MODELS EITHER, SINCE `GEMMA2:2B` (INTERLEAVED LOCAL/GLOBAL SLIDING WINDOW + QK SOFT-CAPPING) IS A MATERIALLY DIFFERENT ATTENTION DESIGN AND STILL RUNS $b>c$ AT A LOOSER ≥ 0.55 CUTOFF ON A LARGER SLICE (n=74). THE FINAL TWO ROWS ARE THE *superseded* SUBGROUP, WHICH FILTERED ON $\geq$ 7B *and* ON $b>c$ AT TEMPERATURE 0; IT IS KEPT FOR PROVENANCE ONLY, IS NOT THE REPORTED SENSITIVITY ANALYSIS, AND IS NOT USED FOR ANY CLAIM; ITS CRITERION READS THE OUTCOME, WHICH IS EXACTLY WHY IT WAS WITHDRAWN. VALUES ARE RECOMPUTED FROM THE PASSAGE-DANGER RUN FILES NAMED IN TABLE S2 BY `ANALYZE_POOLED.PY` (`ALL` AND `SIZE` MODES) AND `ANALYZE_CAPABILITY.PY`.

Pool / probe	stat.	value	reading
MH, 14 strata (primary)	p	1.32×10^{-5}	T=18.99
MH, 7 indep. t0	p	9.72×10^{-4}	T=10.88
MH, 12 size-only	p	2.52×10^{-6}	T=22.15
6 indep. t0	p	4.64×10^{-4}	T=12.25
dilution $12 \to 14$	T	$22.15 \to 18.99$	survives
bootstrap, primary	D	0.0782	652 rows / 47 questions
CI / $P(D \leq 0)$		$[+.003, +.163]$ / .021	clear of 0
bootstrap, size-only	D	0.0860	558 rows
CI / $P(D \leq 0)$		$[+.014, +.168]$ / .009	clear of 0, floor not
gemma2 ≥ 0.55 t0	b/c	12/10	$0.338 \to 0.311$
gemma2 ≥ 0.55 t.6	b/c	13/12	$0.324 \to 0.311$
superseded outcome-filtered pool (10 strata)	T	24.00	p=9.6×10^{-7}, withdrawn
its 5 indep. t0	p	2.66×10^{-4}	T=13.30
its bootstrap	D	0.1034	464 rows, CI $[+.026, +.192]$

TABLE S12

IS THE BENEFIT SIZE A FUNCTION OF MODEL CAPABILITY? (C4, EXTERNAL ANCHOR TEST.) ONE ROW PER MODEL, TEMPERATURE-0 STRATUM ONLY, SO THE SEVEN OBSERVATIONS ARE MUTUALLY INDEPENDENT; D IS THE QUESTION-CLUSTER BOOTSTRAP RISK DIFFERENCE (NAIVE MINUS M-A SUBSTITUTION) ON THE ≥ 0.60 DANGER SLICE. THE ANCHOR IS THE CHECKPOINT SIZE CLASS AS THE TAG AND MODEL CARD REPORT IT, WHICH WE CAN CHECK OFFLINE; WE DELIBERATELY DO NOT BUILD A CROSS-ERA LEADERBOARD ANCHOR, SINCE THESE CHECKPOINTS WERE SCORED UNDER DIFFERENT PUBLIC PROTOCOLS AND ANY SINGLE NUMBER QUOTED WOULD BE ONE WE HAD SELECTED. COCHRAN'S Q TESTS WHETHER THE SPREAD OF D EXCEEDS SAMPLING NOISE; THE SPEARMAN TEST IS EXACT OVER ALL $7! = 5040$ RANK ASSIGNMENTS. NEITHER SUPPORTS A CAPABILITY LAW, SO THE CLAIM WE MAKE IS DIRECTION-CONSISTENCY WITH HETEROGENEOUS, UNDIFFERENTIATED MAGNITUDE. COMPUTED BY `ANALYZE_CAPABILITY.PY` FROM THE PASSAGE-DANGER RUN FILES.

Model	size	D	95% CI	b/c
`gemma2:2b`	2B	+0.043	$[-.128, +.213]$	9/7
`mistral`	7B	+0.149	$[.000, +.298]$	10/3
`llama3`	8B	+0.085	$[-.043, +.213]$	7/3
`qwen3:8b`	8B	+0.000	$[-.085, +.085]$	2/2
`qwen2.5:14b`	14B	+0.128	$[+.043, +.234]$	6/0
`qwen3:14b`	14B	+0.106	$[.000, +.234]$	7/2
`qwen3.8:27b`	27B	+0.068	$[-.068, +.205]$	6/3

heterogeneity: Cochran Q=5.54, 6 df, p=0.48, I^2=0%
anchor: Spearman $\rho(D, \text{size})$=+0.11, exact p=0.82
leave-one-model-out $\rho \in [-0.29, +0.50]$; range of D: 0.149

TABLE S13

SINGLE-FIELD (MTU) DANGER SLICE: THE NULL CONTRAST FOR C3/C4. ON EASY SINGLE-FIELD RETRIEVAL THE HAZARD IS ABSENT ($b = c = 0$ IN EVERY CONFIG, CONTAMINATION 0.000 IN BOTH ARMS, FIDELITY FLOOR INTACT). THE HAZARD CLAIM (C3) IS THEREFORE SCOPED TO NEAR-TWIN-RICH PASSAGE LAYOUTS. RUN FILES: TABLE S2, ROW "FIELD DANGER".

Model	t	LO subst	M-A subst	McNemar p
qwen2.5:14b	0.0	0.000	0.000	1.0
qwen3:14b	0.0	0.000	0.000	1.0
qwen3:8b	0.0	0.000	0.000	1.0
qwen3:8b	0.7	0.000	0.000	1.0

TABLE S14

LAYOUT-ONLY FACTORIAL (LABELS × PREFIX STABILITY) AT FIXED CONTENT, `QWEN3:14B`, TEMP 0, 408 PAIRED TURN-CELLS PER CELL. FRONT REUSE IS MEASURED ON THE CONSTRUCTED PROMPTS WITHOUT ANY MODEL CALL. "OTHER" IS NEITHER GOLD NOR COMPETITOR NAMED. THE LOGIT INTERACTION IS NOT ESTIMABLE (BOTH STABLE CELLS ARE 0/408), SO THE MODEL ROW IS A CLUSTER-ROBUST LINEAR-PROBABILITY FIT.

cell	labels / stable	subst [95% CI]	corr	other	recomp	reuse
s0l0	none / sliding	0.473 [.425, .522]	0.113	0.414	538.2	2.0%
s0l1	named / sliding	0.387 [.341, .435]	0.191	0.422	535.8	7.1%
s1l0	none / stable	0.000 [.000, .009]	0.828	0.172	269.3	51.2%
s1l1	named / stable	0.000 [.000, .009]	0.608	0.392	275.3	51.3%

label effect: stable $+0.000$ (p=1), sliding -0.086 (p=0.006)
stability effect: -0.387 / -0.473; recompute -260.4 / -268.9 tok
LPM: labels -0.086 (p=0.081), stable -0.473 (p=1.3×10^{-8})

TABLE S15

CLAIM LEDGER. GRADE SETS WORDING (STRONG: DEMONSTRATE; MODERATE: INDICATE; NULL: NO-DIFFERENCE). EVERY ROW IS FALSIFIABLE BY THE NAMED EXPERIMENT.

ID	Gr.	Carrying evidence	Falsifier
C1	—	formulation couples F, B, R; constraint binds when edits hit the front	ℓ_t never binding
C2	str.	recompute $1557 \to 249$ ($6.26\times$, CI $[6.22, 6.31]$); pinned recovers 97%	size control closes gap
C3	mod.	max-passage-similarity ROC AUC 0.74–0.81, monotone in the threshold, FDR p=0.001, n=270 trials with 13 events; penalised refit β_{sim}=1.273 (the unpenalised 8.16 sits in near-separation and is not the effect size); null on single-field	sim term $\to 0$ on natural fields
C4	mod.	pre-specified pool of all 14 strata (declared at revision): D=0.0782, CI $[+.003, +.163]$, strict $12/14$ direction	CI spans 0 at larger n
C5	null	joint−prefix: subst. $\Delta-0.0025$, recom. $\Delta-0.22$ tok; TOST inside $\pm 0.03/\pm 20$ margins ($p < 2\times10^{-4}$)	joint beats decoupled $>$ SESOI
C6	str.	head-internal position gradient on the published shared-key stress: correct $0.933 \to 0.008$ front→back of a 24-line head ($\Delta-0.925$, $p=8\times10^{-34}$), $0.475 \to 0.000$ on `llama3`; an append-only head offers no cheap handle on it	gradient absent inside the head, or carried by the volatile tail
C7	mixed	rewritable `INDEX` on the re-keyed grid, read as *exact-match correctness*: long-head middle−front $-0.375 \to -0.158$, DiD $+0.217$ ($p=4.3\times10^{-6}$), placebo difference $+0.400$ at the back, premium 7.40 recomputed tokens per row (CI $[7.19, 7.62]$) with substitution 0.000 throughout. **P2 of the pre-declared reading fails** (residual gradient significant, $p=6.6\times10^{-5}$) and P1 fails on `llama3`. Served again on the shared-key `STATE` (distractor re-keyed `TARGET`, a stale value competing under one key), the block recovers the collapse (index−plain correct $+0.479/+0.662$ at the collapsed slots, qwen3/llama3) and a stale index naming the competitor drives substitution above raw state, so the benefit is not a re-keying artifact (Table S20). Scope: a two-row block in which no key is ever re-bound, so the premium is a whole-block-rewrite upper bound (T7), and head length is fixed within the cost stream, so history-length flatness is the separate Sec. V-B leg rather than a jointly varied factor	block flattens the gradient within arm, or the premium scales with head length rather than rows
C8	mod.	external content rule at the same 552 cells: LLMLingua-2 pruning taken to the view's own measured budget recomputes $+170.1$ tokens/turn (CI $[165.9, 173.8]$) at statistically equal B (-2.9, CI $[-8.7, +2.8]$) and equal F (0.998 vs 1.000, ΔF CI $[-0.005, 0.000]$); unprotected, it keeps the binding in $39/552$ contexts and F falls to 0.071. Two disclosed protocol amendments: the absolute-R-vs-naive contrast was dropped as confounded by B, and the matched rate .20 was chosen from a fit on partial serving of this very leg (the arm actually served is reported, not the fit's pick)	a content rule at matched B reaching the view's R, or the B interval excluding zero (the arms would not be matched)

TABLE S16

POST-HOC SPLIT OF EVERY ANSWER BY FORM (EXPLORATORY: NO PRE-DECLARED CONTRAST, ARM ASSIGNMENT OR TABLE CELL DEPENDS ON IT). THE LAST ROW OF EACH BLOCK IS THE PUBLISHED SHARED-KEY STRESS OF TABLE IV, SCORED WITH THE SAME BUCKETS.

	arm	corr	gold in prose	other record	subst	neither	ambiguous	no-rec	n
qwen3:14b	`plain`	0.753	0.237	0.000	0.000	0.010	0.000	0.000	720
	`index`	0.928	0.069	0.000	0.000	0.003	0.000	0.000	720
	`index_nok`	0.771	0.225	0.000	0.000	0.004	0.000	0.000	720
	shared-key head	0.343	0.025	0.076	0.449	0.104	0.003	0.000	720
llama3:latest	`plain`	0.981	0.001	0.000	0.000	0.018	0.000	0.000	720
	`index`	1.000	0.000	0.000	0.000	0.000	0.000	0.000	720
	`index_nok`	0.979	0.004	0.000	0.000	0.017	0.000	0.000	720
	shared-key head	0.143	0.024	0.011	0.158	0.660	0.004	0.000	720

TABLE S17

WHICH OF THE THREE AXES EACH LINE OF PRIOR WORK TREATS AS ITS OBJECTIVE (QUALITATIVE; "= OBJ." = THE AXIS THE WORK OPTIMIZES, "ADJ." AN ADJACENT EFFECT IT REPORTS BUT DOES NOT OPTIMIZE, "—" OUT OF SCOPE). NO PRIOR BUCKET MAKES BYTE-LEVEL PREFIX STABILITY R A FIRST-CLASS OBJECTIVE OF THE *history-maintenance* DECISION, WHICH IS THE SLOT THIS PAPER FILLS.

Line of work	Fidelity F	Bloat B	Recompute R
Distraction / lost-in-mid	= obj.	—	—
Prompt compression	= obj.	= obj.	adj.
Agent context mgmt.	= obj.	= obj.	adj.
RAG robustness	= obj.	—	—
KV-cache serving	—	adj.	= obj.
View maintenance	—	—	= obj.
This paper	= obj.	= obj.	= obj.

TABLE S18

SYMBOLS. F, B, R ARE THE THREE AXES OF THE TENSION; π RANGES OVER MAINTENANCE POLICIES AND $\mathcal{C}$ IS THE ADMISSIBLE SET UNDER THE PREFIX-STABILITY CONSTRAINT.

Symbol	**Meaning**
H_t	history (turn stream) available at turn t
P_t	rendered prompt/context at turn t
π	maintenance policy, $P_t = \text{render}_\pi(H_t)$
$F_{\text{subst}}(\pi)$	fidelity: 1− substitution rate (primary, forced-commit)
$F_{\text{acc}}(\pi)$	fidelity: exact-match accuracy (format-sensitive; Table II)
$B(\pi)$	bloat: tokens sent per turn ($\lvert P_t \rvert$)
$R(\pi)$	recompute: non-reused prefix tokens per turn
ℓ_t	length of the leading bytes reusable at turn t
$\mathcal{C}$	policies whose head grows append-only
V	volatile-tail size (records kept raw, recent)
B_{KV}	context-window budget in tokens (8192 here)
Δ	tokens in one digest (head) line
t^*	turn at which an unbounded head first meets B_{KV}
$H_{\max}$	cap on head lines under the bounded (folding) policy
m, $\bar{f}$	fold period, realized fold frequency over the analyzed window
I, $I_{\max}$	current-binding index block, and its width cap
Δ_{idx}	tokens per `INDEX` row (7.40 measured, Table S21)

TABLE S19

QUALITY BENEFIT OF THE REWRITABLE INDEX VIEW (SEC. VI-D): EACH CELL IS CORRECTNESS/SUBSTITUTION WITH THE WILSON INTERVAL ON CORRECTNESS; THE QUERIED BINDING SITS AT THE FRONT, IN THE MIDDLE, OR AT THE BACK OF A HEAD OF H LINES. EACH OF THE 18 CELLS HOLDS 120 SESSIONS PER ARCHITECTURE. SUBSTITUTION IS 0.000 IN EVERY CELL OF THIS GRID, AGAINST 0.550–0.742 IN THE PUBLISHED SHARED-KEY STRESS (TABLE IV); TABLE S16 REPORTS WHERE THE LOSS LANDS INSTEAD.

		plain		index		index_nok	
H	slot	qwen3	llama3	qwen3	llama3	qwen3	llama3
1@							
6	front	0.975 / 0.000 [0.929, 0.991]	1.000 / 0.000 [0.969, 1.000]	0.942 / 0.000 [0.884, 0.971]	1.000 / 0.000 [0.969, 1.000]	0.925 / 0.000 [0.864, 0.960]	1.000 / 0.000 [0.969, 1.000]
	middle	0.758 / 0.000 [0.674, 0.826]	0.992 / 0.000 [0.954, 0.999]	0.892 / 0.000 [0.823, 0.936]	1.000 / 0.000 [0.969, 1.000]	0.775 / 0.000 [0.692, 0.841]	0.983 / 0.000 [0.941, 0.995]
	back	0.725 / 0.000 [0.639, 0.797]	0.975 / 0.000 [0.929, 0.991]	0.975 / 0.000 [0.929, 0.991]	1.000 / 0.000 [0.969, 1.000]	0.742 / 0.000 [0.657, 0.812]	0.983 / 0.000 [0.941, 0.995]
1@							
24	front	0.975 / 0.000 [0.929, 0.991]	1.000 / 0.000 [0.969, 1.000]	0.958 / 0.000 [0.906, 0.982]	1.000 / 0.000 [0.969, 1.000]	0.958 / 0.000 [0.906, 0.982]	1.000 / 0.000 [0.969, 1.000]
	middle	0.600 / 0.000 [0.511, 0.683]	0.967 / 0.000 [0.917, 0.987]	0.800 / 0.000 [0.720, 0.862]	1.000 / 0.000 [0.969, 1.000]	0.625 / 0.000 [0.536, 0.706]	0.925 / 0.000 [0.864, 0.960]
	back	0.483 / 0.000 [0.396, 0.572]	0.950 / 0.000 [0.895, 0.977]	1.000 / 0.000 [0.969, 1.000]	1.000 / 0.000 [0.969, 1.000]	0.600 / 0.000 [0.511, 0.683]	0.983 / 0.000 [0.941, 0.995]

TABLE S20

THE SAME CURRENT-BINDING INDEX BLOCK SERVED UNDER *shared-key* HAZARD: THE QUERIED BINDING AND A COMPETING STALE VALUE BOTH KEYED TARGET IN THE PUBLISHED C7 STATE (SEC. S.N). SAME CELL FORMAT AS TABLE S19. EACH OF THE 18 CELLS HOLDS 120 SESSIONS PER ARCHITECTURE; THE STATE IS THE PUBLISHED C7 SHARED-KEY CELL BYTE-FOR-BYTE, SO ONLY THE ONE INDEX ROW SEPARATES THE ARMS. A CORRECT INDEX RECOVERS THE COLLAPSE (Q-A, MEAN INDEX−PLAIN CORRECT RD OVER THE MIDDLE AND BACK SLOTS $+0.479$ QWEN3, $+0.662$ LLAMA3); A STALE ONE (INDEX_NOK, NAMING THE COMPETITOR) DRIVES SUBSTITUTION ABOVE THE RAW STATE (Q-B, MEAN RD $+0.217$ / $+0.590$), I.E. THE MODEL READS THE VIEW OVER THE LINE BENEATH IT. THE BLOCK'S VALUE, NOT ITS PRESENCE, CARRIES THE EFFECT: INDEX−INDEX_NOK CORRECTNESS AVERAGES $+0.632$ / $+0.554$.

		plain		index		index_nok	
H	slot	qwen3	llama3	qwen3	llama3	qwen3	llama3
1@							
6	front	0.942 / 0.000 [0.884, 0.971]	0.375 / 0.000 [0.294, 0.464]	0.958 / 0.000 [0.906, 0.982]	1.000 / 0.000 [0.969, 1.000]	0.058 / 0.817 [0.029, 0.116]	0.717 / 0.258 [0.630, 0.790]
	middle	0.042 / 0.550 [0.018, 0.094]	0.008 / 0.158 [0.001, 0.046]	0.475 / 0.000 [0.388, 0.564]	0.833 / 0.158 [0.757, 0.889]	0.000 / 0.508 [0.000, 0.031]	0.000 / 1.000 [0.000, 0.031]
	back	0.108 / 0.658 [0.064, 0.177]	0.000 / 0.233 [0.000, 0.031]	0.633 / 0.000 [0.544, 0.714]	0.592 / 0.392 [0.502, 0.675]	0.000 / 0.642 [0.000, 0.031]	0.000 / 1.000 [0.000, 0.031]
1@							
24	front	0.933 / 0.000 [0.874, 0.966]	0.475 / 0.000 [0.388, 0.564]	0.967 / 0.000 [0.917, 0.987]	0.983 / 0.000 [0.941, 0.995]	0.175 / 0.683 [0.117, 0.253]	0.600 / 0.292 [0.511, 0.683]
	middle	0.008 / 0.742 [0.001, 0.046]	0.000 / 0.325 [0.000, 0.031]	0.467 / 0.008 [0.380, 0.556]	0.625 / 0.275 [0.536, 0.706]	0.000 / 0.667 [0.000, 0.031]	0.000 / 0.983 [0.000, 0.031]
	back	0.025 / 0.742 [0.009, 0.071]	0.000 / 0.233 [0.000, 0.031]	0.525 / 0.017 [0.436, 0.612]	0.608 / 0.242 [0.519, 0.691]	0.000 / 0.675 [0.000, 0.031]	0.000 / 0.958 [0.000, 0.031]

TABLE S21

COST OF THE SAME VIEW UNDER MATCHED CONDITIONS (SEC. VI-D2): THE INDEX BLOCK OF I CURRENT BINDINGS REPLAYED IN THE PUBLISHED COST STREAMS. PREMIUM IS THE PAIRED RECOMPUTE DIFFERENCE AGAINST PREFIX_STABLE WITH A SESSION-CLUSTER BOOTSTRAP INTERVAL. CLUSTER-ROBUST SLOPE OF THE PREMIUM ON THE NUMBER OF INDEX ROWS: 7.40 TOKENS PER ROW (SE 0.11, $p = 3.55 \times 10^{-24}$), INTERCEPT 21.1 (SE 0.18), OVER 1,380 OBSERVATIONS (3 INDEX SIZES $\times$ 460 PAIRED TURNS) IN 20 SESSIONS; THE FITTED LINE IS WITHIN 11.4 TOKENS OF EVERY MEASURED LEVEL. INFERENCE BASIS: THE SAME REGRESSION WITH A SESSION-CLUSTERED SANDWICH COVARIANCE, t ON 19 DF; THE $[7.19, 7.62]$ INTERVAL THE INTRODUCTION QUOTES FOR THE SLOPE IS THE WALD ± 1.96 SE OF THIS FIT, WHEREAS THE PREMIUM INTERVALS IN THE COLUMNS ARE 2.5/97.5 PERCENTILE SESSION-CLUSTER BOOTSTRAPS. IN THE SAME STREAMS LENGTH_ONLY RECOMPUTES $1,566.8$ AND PREFIX_STABLE 250.8 TOKENS PER TURN.

arm	calls	prompt	cache	recompute	premium vs stable [95% CI]	prefill ms	corr
prefix_stable	460	453	44.6%	250.8	—	212.1	1.000
index_I2	460	481	42.1%	278.5	27.7 / [27.1, 28.2]	240.1	1.000
index_I8	460	545	37.1%	342.6	91.8 / [89.5, 93.9]	278.2	1.000
index_I24	460	650	31.3%	446.6	195.7 / [190.3, 200.7]	329.4	1.000
length_only	460	1,591	1.5%	1,566.8	—	1,039.7	1.000